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Review

Comprehensive Review of Edge Computing for Power Systems: State of the Art, Architecture, and Applications

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Abstract: The increasing complexity of conventional energy distribution systems, combined with the growing demand for efficient data processing, has necessitated the implementation of smart grid technologies and the integration of advanced computing paradigms such as edge computing. Traditional cloud-based solutions face significant challenges, including high latency, limited bandwidth, and cybersecurity vulnerabilities, rendering them less effective for real-time smart grid applications. Edge computing enables localized data processing, which significantly reduces latency and optimizes bandwidth usage. These capabilities enhance the resilience and intelligence of modern energy systems. This paper presents a systematic review of edge computing in energy distribution systems, examining its architectures, methodologies, and real-world applications. Key application areas consist of real-time data transmission, smart metering, microgrid management, anomaly and fault detection, state estimation, and energy management. The analysis shows how edge computing improves secure communication, supports decentralized intelligence, and facilitates scalable energy optimization. Beyond these advantages, the review also identifies critical challenges such as interoperability issues, resource constraints, and security vulnerabilities. By categorizing edge computing applications, the findings provide a comprehensive reference for both researchers and industry professionals working on the development of next-generation energy management systems.

Keywords: edge computing; fog computing; cloud computing; smart grid; power system application

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1. Introduction

The increasing demand for smart energy systems has led to a rapid rise in the number and variety of Internet of Things (IoT)-based smart devices in power systems [1]. IoT has emerged as a key enabler of smart grids, facilitating real-time monitoring, predictive maintenance, and decentralized control of energy systems [2]. This technological transformation not only enhances system efficiency but also lays the foundation for more resilient and adaptable grids, which are critical for the future of energy management. However, the exponential growth in the number of IoT devices has resulted in massive data generation, necessitating high bandwidth and powerful data processing infrastructures. Projections made over a decade ago foreshadowed this trend: Cisco Internet Business Solutions Group (IBSG) predicted that the number of internet-connected smart devices would reach approximately 25 billion by 2015 and 50 billion by 2020 [3]. According to the International Data Corporation (IDC), the amount of digital data generated in 2010 exceeded 1 zettabyte, and

as of 2012, approximately 2.5 exabytes of data are generated every day [4]. Dell Technologies estimated that by 2025, there would be 41.6 billion IoT devices worldwide, generating a total of 79.4 zettabytes of data [5]. This exponential data growth calls for innovative solutions to handle the surge in data traffic and to ensure the effective management of network resources.

The increasing volume of data generated by IoT devices necessitates advanced big data analytics and processing capabilities. Cisco's 2018–2023 report highlights the urgent need for innovative solutions in network architecture and data management to address the challenges posed by this data explosion [6]. This underscores the importance of developing efficient network infrastructures and management frameworks capable of accommodating the exponential growth in connected devices and data traffic.

In response to the increasing demands of modern data management, cloud computing (CC) emerged as a significant technological advancement. Introduced by IBM and Google in 2007, CC provided new methodologies for software development, particularly in addressing the challenges of internet-scale applications. These technologies offered scalability, flexibility, and distributed data management, enabling businesses to manage their data without the need for on-premises infrastructure [7]. CC provides flexibility and scalability in storing, accessing, and processing data. This technology is seen as a unique solution for delivering applications to businesses [8]. However, while CC offered scalable solutions, it also introduced new challenges, particularly in the area of security. As cloud providers increasingly host vast amounts of sensitive data and core business applications, ensuring the confidentiality, integrity, and security of these data have become a pressing issue [9]. Today, security and privacy issues have emerged as major challenges when cloud providers hosting large datasets and core applications share these data with their customers [10]. Thus, although CC has provided a flexible and scalable solution for businesses, its rapid adoption has also highlighted the need for robust security frameworks to address the growing risks associated with cloud environments. The rapid expansion of the IoT has led to an unprecedented surge in data generation at the network edge. Traditional CC architectures struggle to efficiently process this influx of information while maintaining low latency and high security. Relying solely on centralized cloud processing increases response times, exacerbates bandwidth congestion, and elevates privacy risks. To address these limitations, fog computing (FC) emerged as an intermediate solution that reduces network traffic and provides storage and computation near the edge of the network [11]. Existing solutions such as FC attempt to bridge this gap, yet they remain insufficient for real-time, latency-sensitive applications [12]. To overcome these challenges, EC has emerged as a more effective approach by enabling localized data processing at the source. Unlike CC, EC minimizes communication latency by reducing the distance data must travel before being analyzed, resulting in significantly lower response times [13]. At this point, EC significantly reduces communication latency compared to the cloud by processing data at the source. Time-sensitive data analysis tasks can be processed at the edge to reduce the latency caused by data transfer [14]. Still, EC not only provides the advantage of low latency but also has the potential to solve energy consumption, bandwidth overhead, and security issues. This localized processing reduces the latency associated with data transfer and offers distinct advantages for time-sensitive applications. EC not only provides the benefit of low latency but also has the potential to address issues related to energy consumption, bandwidth overhead, and security [15].

Each of the existing works in the literature focuses on different aspects of the applications of EC in smart grids, often delving into specific areas. For instance, ref. [16] provides an overview of the potential of EC in the SG, noting that only a limited number of studies have directly applied EC to SG. Ref. [17] provides a theoretical framework, concentrating

on the hardware and software components of the EC-CC information architecture, but with limited coverage of practical application scenarios. Ref. [18] provides an overview of IoT and edge computing-based smart grid systems, addressing their architecture, integration with SCADA systems, and key challenges such as scalability, sustainability, security, and resilience. In this context, it focuses on key application areas such as power distribution monitoring, microgrids, and advanced metering systems. Similarly, ref. [19] proposes solutions for IoT-based smart grids where cloud computing is insufficient, such as real-time data processing, terminal management, and privacy protection. In the reviews on fog computing and related paradigms (including EC) [20], research topics have been categorized and classified. Ref. [21] introduces the emergence and capabilities of EC, examines state-of-the-art architectural developments, and explores communication techniques.

This paper synthesizes the fragmented nature of the existing literature, systematically analyzing the relationships between EC, FC, and CC paradigms in power systems, both theoretically and practically. It spans all layers of power systems, from generation to consumption, from transmission to load management, from security to data privacy, and from field applications to energy optimization. The role of information technology (IT) paradigms, previously addressed in a fragmented manner in the literature, is integrated with an energy systems perspective. Therefore, this paper provides a unique and in-depth contribution to the literature by offering a comprehensive, multi-layered, and comparative overview, including application scenarios, advantage-disadvantage analyses, and future research directions. By performing so, it establishes a structured framework for evaluating the impact of EC on modern power systems, considering various computational methodologies, technological advancements, hardware and software components, architectural approaches, and application domains. A systematic literature review was conducted, encompassing over 500 publications. From these, 237 relevant and cohesive studies were selected for detailed analysis. These included peer-reviewed research articles, conference proceedings, and technical reports. The sources were drawn from a variety of applied science journals, including those focused on energy systems, computer science, and electrical engineering. The review also incorporated publications from prominent energy systems conferences related to smart grids, power distribution, and renewable energy technologies.

The main contributions of this study are as follows:

- Systematic Literature Review: A systematic review is conducted to analyze the role of EC in energy distribution systems, addressing its potential advantages and limitations in comparison to other computing paradigms.
- Technical Comparison of Computing Architectures: EC is critically compared with CC and FC in terms of essential criteria such as latency, computational efficiency, data transmission, and security, emphasizing their contextual applicability in power systems.
- Real-Time Data Processing and Optimization Techniques: The integration of EC with communication protocols such as Minimum Message Queuing Telemetry Transport (MQTT) and Advanced Message Queuing Protocol (AMQP), container technologies like Docker and Kubernetes, and AI-driven predictive maintenance models is explored. The study also analyzes the role of EC in sensor data processing to enhance fault prediction accuracy and optimize demand-side management in energy systems.
- Technological Components and Infrastructure: The role of hardware and software architectures in EC-enabled energy systems is analyzed. The study explores computational frameworks, edge devices, virtualization technologies, and distributed resource allocation strategies that contribute to the efficient deployment of EC.
- Applications in Power Systems: The advantages of EC in various power system applications, including data transmission, real-time monitoring and control, microgrid

systems, smart metering, anomaly detection, state estimation, energy management, and fault detection, are assessed in terms of their contributions to grid stability, operational efficiency, and system reliability. The study evaluates how EC-based solutions improve energy consumption modeling, enable decentralized energy management, optimize grid stability, and enhance system reliability.

- Security and Privacy Solutions: By identifying major vulnerabilities in EC-powered energy systems, the paper proposes mitigation strategies including decentralized authentication, hybrid encryption, and secure transmission protocols to strengthen system resilience.
- Future Research Directions: Key areas requiring further investigation are highlighted, including scalable EC architectures, energy-efficient edge processors, advanced data management techniques, and the integration of distributed AI applications to enhance the resilience and efficiency of energy systems.

The structure of this paper is illustrated in Figure 1 and organized as follows. Section 2 presents an overview of various computing paradigms utilized in power systems, including CC, EC, FC, Mobile Cloud Computing (MCC), and Mobile Edge Computing (MEC). It also presents a comparative analysis of these paradigms, highlighting their respective strengths and limitations in the context of energy distribution systems. Section 3 examines the core technologies, hardware, and software tools employed in EC, with subsections dedicated to data processing techniques, container technologies, communication protocols, and the software platforms and hardware tools that facilitate EC deployment. Section 4 discusses the diverse applications of EC systems, such as data transmission, monitoring and control, microgrid systems, smart metering, anomaly detection, state estimation, energy management, and fault detection. In Section 5, security and privacy concerns associated with EC are addressed, and potential solutions are proposed. Section 6 identifies the challenges faced by EC systems and outlines potential future research directions to advance the field. Finally, Section 7 concludes the paper by summarizing the key findings and contributions.

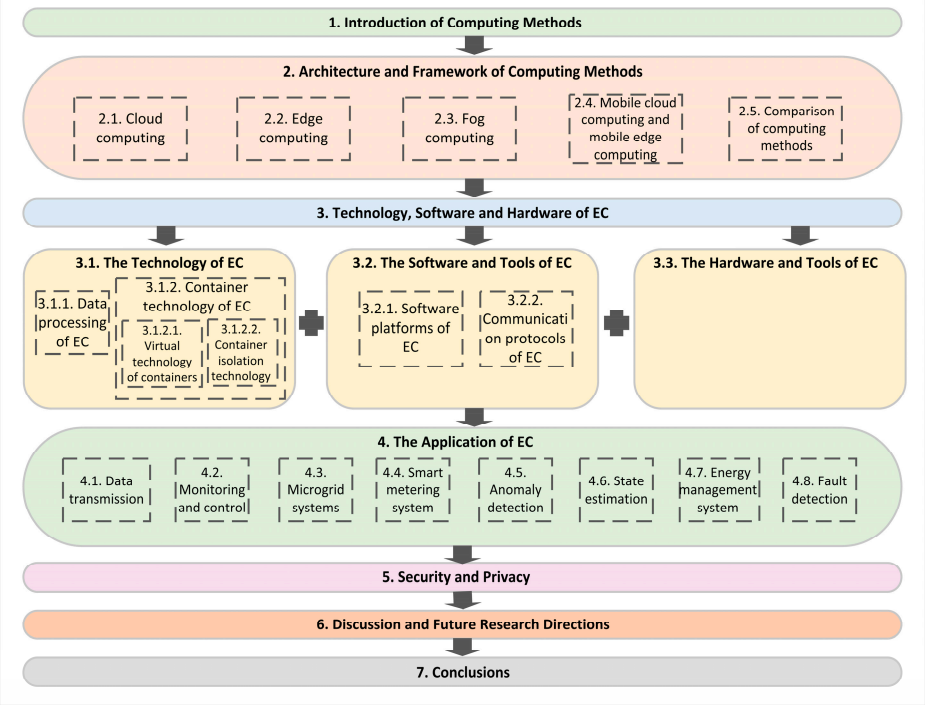


Figure 1. Structure of this paper.

2. Architecture and Framework of Computing Methods

2.1. Cloud Computing

CC refers to the delivery of storage, services, and application resources over the Internet [22]. The National Institute of Standards and Technology (NIST) defines CC as "a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction" [23]. This model offers significant advantages, including cost efficiency, resource scalability, and broad accessibility. However, despite these benefits, CC also presents several critical challenges. Its widespread and centralized nature increases its vulnerability to cyber threats, making it an attractive target for various forms of cyberattacks [24]. Issues such as data breaches, insecure APIs, and shared vulnerabilities are particularly concerning in applications that handle sensitive information [9].

Another fundamental limitation of CC lies in its insufficient support for latency-sensitive applications, particularly within IoT environments. Although cloud data centers are adept at processing large volumes of IoT-generated data efficiently, their centralized architecture often introduces latency, which impedes real-time responsiveness. While some studies have explored optimization strategies to reduce these delays in CC-based systems, others argue that inherent structural limitations hinder their ability to meet stringent real-time requirements [25]. This limitation is particularly critical in power management systems, where timely, data-driven decisions are essential for maintaining stability and reliability [26]. For instance, cloud-based implementations using heuristic techniques such as genetic algorithms (GA) have been developed to schedule IoT requests and minimize overall latency [27]. However, these solutions frequently fail to meet the stringent real-time performance requirements. As a result, emerging paradigms like FC and EC have gained traction for their ability to process data closer to the source, thereby reducing response times and enhancing system performance [28].

Despite its latency challenges, CC remains integral to supporting the self-healing capabilities of smart grids, thereby contributing to energy savings and enhancing operational efficiency [29]. Its integration has become increasingly important with the growing deployment of renewable energy resources, as CC enhances both system stability and reliability [30]. By optimizing resource allocation through techniques such as time-sharing of computing cores, CC facilitates cost-effective and efficient computational processes across multiple devices [31]. Moreover, CC-based simulation platforms offer practical and economical solutions for power system analysis and decision-making support. For instance, in [32], a cloud-based framework is presented that monitors and controls power generation and consumption to minimize energy waste. CC-based optimization systems also play a critical role in extending the operational life of hybrid renewable energy systems and battery storage technologies, enabling more flexible and adaptive power distribution models [33]. The advancements in CC also improve storage performance and adaptive control mechanisms in energy management systems.

Despite these advantages, challenges persist. The efficient utilization of renewable energy resources in smart grids is hindered by issues related to big data management and the intermittent nature of renewable energy sources. To address these challenges, ref. [34] highlights the necessity of incorporating CC into smart grid energy management, emphasizing its role in handling large-scale data processing and decision-making. A specific model, CloudIoT, has been introduced to address the complexity of managing vast data streams and user demands in IoT-integrated smart networks [35]. This model improves interoperability and facilitates the seamless integration of CC and IoT technologies. Studies

such as [36,37] analyze the synergistic benefits of CC and IoT, demonstrating how this combination enhances performance, scalability, and adaptability in smart grids.

2.2. Edge Computing

EC is a network paradigm that aims to perform computations as close as possible to the data source in order to reduce latency and optimize bandwidth usage. In other words, it refers to offloading certain processes from centralized cloud environments to localized nodes to localized locations, such as a user's computer, an IoT device, or an edge server [38]. More specifically, ref. [12] defines EC as "a new computing model performed at the edge of the network" and encompassing network resources and computing processes between the data source and the CC center. Ref. [39] emphasizes that EC aims to provide more efficient services by placing computing and storage resources closer to mobile devices or sensors at the network edge. Similarly, ref. [40] states that EC delivers real-time services by reducing transmission delays and bandwidth usage with computing and storage resources placed at the edge of the network. This technology enhances data processing and network optimization by providing front-end computational support for applications such as mobile devices and the IoT.

EC refers to the decentralization of storage and computing resources and locating them closer to the data source. In this approach, raw data are analyzed directly near the network edge rather than being sent to a central cloud server for processing and analysis [41]. Moreover, EC refers to the physical computing infrastructure positioned between the device and the hyperscale cloud, enabling support for various applications. In ref. [42], EC provides to provide more efficient and faster services by bringing the capabilities of CC closer to the device, i.e., the data source.

EC is particularly effective in smart grid systems due to its ability to perform computations closer to the data source, reducing latency and optimizing bandwidth usage. It consists of a three-layer architecture: (1) the device layer, including edge devices (2) the edge server layer and (3) the cloud layer [43]. The device layer includes various edge devices with limited computing and storage resources, such as sensors, actuators, and IEDs (e.g., relays, energy analyzers, and quality recorders) located in distribution centers of the power grid, which are responsible for data collection. This layer offers faster and more flexible solutions by processing current, voltage, and fault current data from IEDs when anomaly conditions are detected. The network connecting the device layer to the edge server layer and the edge server to the cloud server layer requires high resolution, high data rates, and high bandwidth for data transmission. Wireless connections are often preferred over wired connections. The edge server layer, positioned between the cloud server layer and the device layer, brings the cloud server tasks closer to the data source, reducing latency and bandwidth usage [21]. The cloud server layer is responsible for further processing and managing large volumes of data that have been pre-processed at the edge server layer. Figure 2a illustrates the general hierarchical architecture of EC.

The literature contains numerous applications of EC across various domains. In Industry 4.0, ref. [44] examines the integration of EC with Smart Manufacturing applications, highlighting its benefits during the digital phase of the Physical-Digital-Physical (PDP) cycle and advancements in Industry 4.0. Ref. [45] explores the use of computational intelligence (CI) techniques to address challenges such as resource allocation and task offloading in both CC and EC, emphasizing the complex nature of these optimization problems, including their convexity and NP-hardness. Furthermore, EC has been proposed as a solution to the big data problem generated by IoT-connected devices by offering distributed computation and storage services, and a distributed multi-level storage (DMLS) model

and multi-factor least utilized (mLFU) algorithm have been used to manage constrained resources [46].

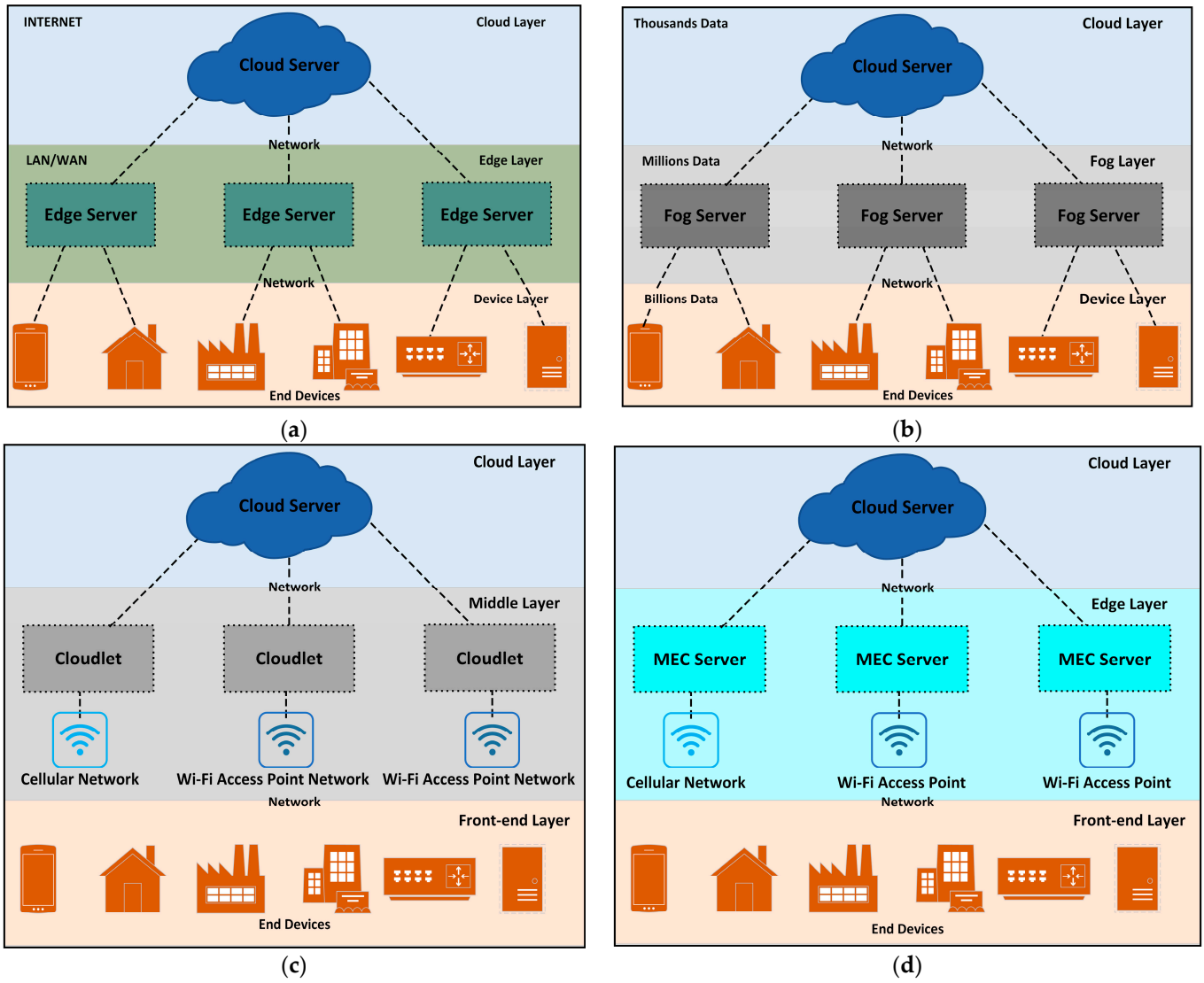


Figure 2. General hierarchical architecture of computing methods. The overall hierarchical architecture of (**a**) EC; (**b**) FC; (**c**) MCC; (**d**) MEC.

In the energy sector, ref. [47] focuses on improving efficiency by analyzing existing architecture and decision-making criteria, utilizing optimization algorithms, and SWOT analysis to support edge offloading and the implementation of AI-enabled energy services in smart grids. Ref. [48] emphasizes that EC focuses on providing services and computations near the source of data generation to meet the needs of industry and real-time applications. Ref. [49] proposes a communication and computation resource allocation model to optimize offloading decisions between multi-access edge computing and cloud servers. This model aims to optimize performance while minimizing resource usage for computation offload requests with low latency requirements. Ref. [50] investigates how EC enhances IoT by bringing data processing and storage closer to the user while simultaneously highlighting significant risks to data security and privacy despite the improved quality of service. Ref. [51] examines EC and edge artificial intelligence (Edge AI) applications, presenting approaches for deploying AI algorithms on edge devices. These studies comprehensively address the potential and challenges of EC in various domains.

2.3. Fog Computing

FC is an architecture designed to extend cloud capabilities by integrating computation, storage, and networking services along a continuum between end devices and centralized cloud data centers [52]. Introduced by Cisco Systems in 2012, FC is described as "an extension of the CC paradigm that provides computation, storage, and networking services between end devices and traditional cloud servers" [53]. Cisco applied this technology by extending the CC paradigm to the edge of networks, specifically wireless networks for the Internet of Things (IoT) [46]. The OpenFog Consortium defines FC as "a horizontal system-level architecture that distributes compute, storage, control, and networking functions closer to users along the continuum from the cloud environment to objects" [54]. For instance, compared to large-scale cloud data centers (e.g., Microsoft Azure regions), an IoT device typically communicates with a fog node that resides in closer proximity [55]. Figure 2b illustrates a three-tier fog system: the cloud layer, the fog layer, and the IoT/end-user layer. The fog layer can consist of one or more fog domains, each managed by different providers. Communication between the IoT/End-user layer and the fog layer occurs over the LAN, while communication with the cloud layer happens over the WAN, sometimes via the fog [56].

FC facilitates a seamless service continuum between cloud infrastructure and end devices, enhancing computing flexibility in heterogeneous environments. These two technologies provide ubiquitous computing, storage, control, and communication capabilities, offering mutually dependent and beneficial services to create a continuous continuum of services. FC is a decentralized computing infrastructure situated between edge devices and the cloud, working with heterogeneous devices. Fog Computing Networks (FCN) support non-IP-based access technologies for communicating with end devices, as well as functions such as resource allocation, security, and device management [57]. By enabling data processing closer to the user, FC minimizes latency and enhances the efficiency of data handling.

To address performance and efficiency requirements in FC environments, ref. [58] introduced a dynamic Integer Linear Programming (ILP) method to optimize resource allocation from the FC layer to IoT devices. This method aims to ensure that tasks are completed within time constraints while minimizing service latency and energy consumption, particularly with devices that have limited processing power and energy capacity. A fog orchestrator is proposed to centrally organize the resource pool, match applications to specific requests, and provide an automated workflow to physical resources [59]. This orchestrator manages deployment and scheduling, performs workload management with runtime quality of service (QoS) control, and aims to manage specific objects efficiently. Resource management remains a challenging problem due to constraints and dynamic workloads. While AI and ML algorithms have been employed to address these challenges, issues such as high variance, explainability, and the difficulty of online training persist [60].

Security and privacy remain critical challenges in FC systems due to their distributed and dynamic architecture. Traditional intrusion detection mechanisms often prove inadequate in such environments. To address this, ref. [61] proposes a distributed deep learning framework designed to detect cyberattacks in edge and fog-based IoT networks. Furthermore, ref. [62] presents FogBus, a framework supporting seamless IoT–Fog–Cloud integration. FogBus supports platform-independent execution for multiple applications and leverages blockchain, encryption, and authentication technologies to ensure security across the continuum.

2.4. Mobile Cloud Computing and Mobile Edge Computing

The IoT enables a wide range of innovative applications aimed at enhancing user experience and quality of life [63]. However, the centralized architecture of CC introduces

several limitations, including constrained bandwidth, increased latency, and higher energy consumption, which impede the fulfillment of real-time performance demands. To overcome these challenges, MEC has emerged as a distributed computing paradigm that brings computational and storage resources closer to end-users by integrating edge capabilities into mobile networks [64]. By embedding EC functionalities into mobile infrastructures, MEC supports real-time responsiveness for latency-sensitive applications and enhances network efficiency [65]. In 2014, the European Telecommunications Standards Institute (ETSI), through its Industry Specification Group (ISG), formalized the MEC architecture to enable seamless convergence of cloud functionalities within mobile network environments.

MCC combines mobile computing and cloud computing paradigms by employing computation offloading techniques to mitigate the inherent resource constraints of mobile devices [66]. Despite offering benefits such as extended battery life, enhanced storage capabilities, scalability, and reliability, MCC still faces significant challenges in terms of security, privacy, trust, bandwidth limitations, data management, synchronization, energy efficiency, and system heterogeneity [67]. To alleviate these concerns, MEC shifts control, storage, and processing operations to the network edge, allowing resource-constrained mobile devices to execute computationally intensive, low-latency applications more effectively [68]. In contrast to MCC, where computational tasks are offloaded to remote cloud servers via the Internet, MEC executes processing tasks closer to the end-user, thereby enhancing both Quality of Experience (QoE) and Quality of Service (QoS). A hierarchical deployment architecture for MCC and MEC is illustrated in Figures 2c and 2d, respectively.

The Introductory Technical White Paper defines MEC as a computing paradigm that allows computationally intensive and latency-sensitive tasks to be executed on mobile devices by leveraging unused processing power and storage capacity at the network edge [69]. In parallel, Cisco introduced FC as a broader concept that extends MEC beyond smartphones to include other edge devices, such as set-top boxes [53].

One of the key applications of MEC is face recognition, which involves five main computational steps: image acquisition, face detection, preprocessing, feature extraction, and classification [70]. Moreover, MEC enhances the efficiency and responsiveness of Augmented Reality (AR) applications on mobile devices [71]. These systems typically comprise five essential components: a video source, tracker, mapper, object recognizer, and renderer. While the video source and renderer are executed locally, the tracker, mapper, and object recognizer can be offloaded to the edge server due to their high computational demand [71]. This offloading strategy minimizes redundant data transmission by aggregating user input in shared geographic regions, thereby improving overall system performance and energy efficiency. This approach allows mobile users to fully leverage MEC's benefits, particularly in reducing latency and improving energy efficiency. A comparison of MEC and MCC systems is provided in Table 1.

Table 1. The differences between MEC and MCC.

	Mobile Edge Computing	**Mobile Cloud Computing**
Distribution/Deployment	Requires simple configuration and planning	Requires complex configuration and planning
Distance to End Users	Close to users	Remote to users
Server Hardware	Compact data centers	Large-scale data centers
Server Location	Wireless gateways	Servers are located in large areas
Use of Backhaul	Relieves network congestion	The risk of network congestion is high
System Management	Centralized and distributed	Centralized
Latency	Very low latency	High latency

2.5. Comparison of Computing Methods

The integration of cloud, fog, edge, and IoT layers forms a multi-tiered architecture, with a central cloud—potentially distributed across various geographies—at the top. Beneath the cloud, multiple fog nodes act as intermediaries, each managing numerous edge devices that, in turn, interface with a wide array of IoT devices [72]. Although EC and FC share overlapping functionalities, their key distinction lies in the scope and location of intelligence deployment. Specifically, FC enables computation directly in IoT devices, thereby aligning more closely with decentralized IoT architectures [42]. Conversely, EC focuses on processing data at or near the data source, thereby reducing latency and alleviating the burden on central cloud infrastructure by transmitting only filtered or aggregated data for deeper analysis. This collaborative ecosystem between CC, FC, and EC fosters efficient, scalable, and resilient solutions across various sectors, including smart manufacturing, energy systems, cybersecurity, and intelligent home automation.

Despite its scalability and processing power, CC faces challenges related to exponential data growth and network bandwidth limitations [73]. By contrast, EC offers superior performance in terms of response time and real-time data processing, making it particularly suitable for time-sensitive applications. However, while edge servers offer better computing and storage capabilities compared to IoT devices, they still lag behind cloud servers in terms of overall computational power. As a result, resource-intensive tasks are often offloaded to cloud infrastructures, which, although providing enhanced processing capabilities, can introduce latency [74]. A comparative overview of CC, FC, and EC is presented in Table 2, outlining their architectural distinctions, latency characteristics, and application domains.

Table 2. The comparison between cloud, fog and edge computing.

	Cloud Computing	Fog Computing	Edge Computing
Location of data collection, processing, storage	A cluster of data center servers hosted on the internet	Near-edge and core networking, network edge devices, and core networking devices	Network edge, edge devices
Computing power	Strong (depend on server cluster)	Weak (depend on network edge device network)	Common (depend on edge device)
Responsible for the type of task	Large computation, or long-term storage task	Preprocessing	Real-time processing
Focus	Clusters level	Infrastructures level	Things level
Handling multiple IoT applications	Supported	Supported	Unsupported
Resource contention	Slight	Sligh	Serious
Latency	High	Low	Ultra-low
Privacy and Security	Low	Low	High

Table 3 presents a detailed comparison of CC, FC, and EC in the context of energy systems, highlighting their differences in latency, energy efficiency, scalability, and real-time processing capabilities. The comparison emphasizes why EC is particularly advantageous for real-time energy management and distributed energy resources (DERs).

Table 3. The comparison between CC, FC, and EC for energy systems.

	Cloud Computing	Fog Computing	Edge Computing
Energy Efficiency	Low (high energy consumption due to data transmission and processing at large-scale data centers)	Medium (optimized for regional energy use)	High (minimizes transmission energy, efficient local processing)

Table 3. *Cont.*

	Cloud Computing	Fog Computing	Edge Computing
Data Transmission Cost	High (large amounts of data sent to cloud centers)	Medium (aggregates and pre-processes data before sending to cloud)	Low (only critical data are sent, reducing bandwidth usage)
Scalability	High (scales based on cloud provider infrastructure)	Medium (limited by regional computing capacity)	High (scales as more edge devices are deployed)
Resilience to Network Failures	Low (dependent on continuous connectivity)	Medium (operates even if the cloud connection is lost)	High (independent operation with local processing)
Uninterrupted Power and Resilience	Requires stable power for data centers; failure may impact multiple applications	Provides flexibility in managing energy at a regional level	Can operate during grid outages, ensuring continuous data processing
Integration with Renewable Energy Sources	Supports large-scale renewable energy integration, but with delays in response	Manages regional renewable energy fluctuations, optimizing local usage	Directly interfaces with local renewable sources like microgrids and DERs
Data Aggregation and Preprocessing	Minimal (raw data sent to the cloud for analysis)	Moderate (aggregates and filters data before sending)	High (processes critical data locally, reducing transmission needs)
Support for Smart Grids	Used for large-scale smart grid analytics and decision-making	Supports regional smart grid operations and energy management	Essential for real-time control of smart grids, managing demand-response programs
Support for Distributed Energy Resources (DERs)	Centralized monitoring and optimization of DERs	Enables coordination of distributed energy assets	Direct integration with DERs for local optimization and real-time adjustments
Support for Demand Response (DR) Programs	Delayed responses due to cloud processing times	Supports aggregated demand response strategies	Real-time, automated demand response capabilities

3. Technology, Hardware and Software of Edge Computing

3.1. The Technology of Edge Computing

EC technologies in power systems are widely used in distributed computing areas such as data transmission and processing. These technologies also reveal the main working principles of the EC architecture.

3.1.1. Data Processing of Edge Computing

CC often struggles to meet the rising computational demands due to its centralized nature. In contrast, EC addresses this challenge by distributing computation, data storage, and processing to local edge nodes, ensuring more efficient data handling. These edge nodes process and cache data locally, predicting user traffic demands, which helps reduce latency by avoiding the need to download information from remote data centers [75]. Moreover, EC significantly reduces the bandwidth burden on centralized clouds, as it manages data locally and performs real-time computations at the edge, minimizing end-to-end latency. For instance, ref. [76] introduces a method to improve video streaming experiences within smart cities by autonomously creating MEC services. This approach enables seamless data access anytime, anywhere, ensuring optimal QoE while reducing latency. Video streams from monitoring devices are processed and analyzed locally on MEC servers, with relevant information forwarded to application servers to further alleviate network traffic. Furthermore, ref. [77] explores the use of serverless computing platforms to address challenges in running data-intensive applications in edge environments. This

approach optimizes computation and data movement by adapting to workload-specific requirements, ensuring efficient use of edge infrastructure.

A growing area of research is the integration of blockchain with EC, primarily to tackle privacy concerns in IoT data processing. To mitigate security risks, specialized EC environments are being developed that continuously monitor and audit node activities. For example, ref. [78] introduces Recordchain, a blockchain architecture tailored to EC environments, addressing bandwidth and security issues effectively. Figure 3 illustrates the data processing flow in EC, where each edge node handles a specific IoT device region, processing data locally and optimizing storage and bandwidth usage. IoT data are categorized as short-term or long-term, with short-term data being discarded once obsolete, ensuring efficient resource utilization.

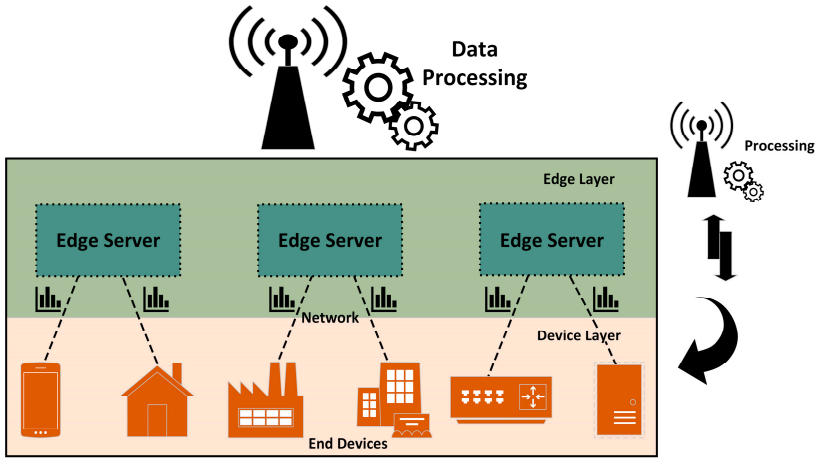


Figure 3. Localized data processing architecture in EC [78].

In efforts to address high transmission delays and limited bandwidth, several studies propose hybrid network architectures that combine CC and EC. For example, ref. [79] introduced a three-layer network architecture designed to reduce communication delays by optimizing device cooperation at the edge layer using Kruskal's algorithm to compute a minimum spanning tree. Additionally, ref. [80] examines performance metrics such as message transmission delays and response times, focusing on large dataset processing within MEC environments. Another notable proposal for smart grid environments [81] integrates P2P technology with EC to enhance data processing and transmission efficiency through distributed data collection, computation, and storage.

Data management in IoT also benefits from EC by enabling the categorization of data into distinct segments for more efficient processing. Ref. [82] introduces such a framework, while [83] proposes a statistical analysis scheme of multidimensional smart grid data, ensuring user privacy preservation. Moreover, ref. [84] presents an EC-based platform for distributed networks, where computational tasks are distributed across edge nodes prior to central data aggregation. The importance of data security in EC is also evident in studies focused on smart grids. For example, ref. [85] introduces a monitoring model for EC-based grids, which facilitates the timely detection of network failures and supports real-time responses.

The transformational impact of EC on operational efficiency, data security, and business optimization is increasingly acknowledged. By reducing latency and improving autonomous decision-making and analytics capabilities [86], EC is reshaping industries. Research on hardware accelerators for EC underscores their role in achieving low latency, low power consumption, and high reliability. Ref. [87] explores the use of the CORDIC algorithm to pre-calculate micro-rotation directions, proposing a multi-stage structure to

enhance EC performance. Overall, these developments demonstrate how EC is reshaping traditional data-handling paradigms by enabling localized, secure, and real-time processing capabilities, particularly in data-intensive and latency-sensitive environments like smart cities and grids.

3.1.2. Container Technology

Container technology is a virtualization method used to increase the portability and efficiency of applications. This technology enables applications to run in an isolated environment with all their dependencies, accelerating software development and deployment processes and optimizing resource usage. Containers are widely preferred in modern infrastructures such as microservice architectures, CC, and EC.

Virtual Technology of Containers

Containers have become a widely adopted method for running applications in EC. However, before an application can be launched, the end node must download the container image, which consists of various layers. Due to the limited bandwidth available in EC environments, the initialization latency due to long download times can significantly impact real-time performance. To mitigate this, ref. [88] proposes a specific sequence for container assignment and layer downloads to minimize initialization latency. While layers in Docker images are not entirely identical, ref. [89] argues that reorganizing the Docker image layers to maximize their overlap can reduce both storage and network consumption. Furthermore, ref. [90] emphasizes that integrating MEC, network function virtualization (NFV), and container-based virtualization technology (CVT) in 5G networks provides low latency, high bandwidth, and agility, creating a flexible and resilient network compatible with IoT devices. In [91], lightweight virtualization technologies are explored in IoT and MEC environments to enhance container efficiency on low-power devices. Additionally, virtualization in Edge/Fog servers supports scalable and high-performance services, as illustrated in Figure 4, which highlights the effectiveness of mechanisms that provide on-demand services by launching customized VMs or containers [92].

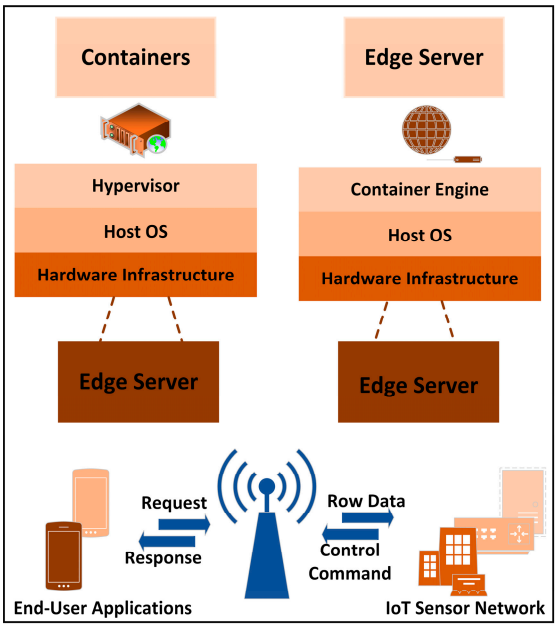


Figure 4. Virtualization layers and components in EC Architecture [92].

Virtualization, or containerization at the operating system level, is commonly used to separate process groups [93]. Prominent projects in this area include Docker (San Francisco,

CA, USA), LXC (Global), CoreOS (San Francisco, CA, USA), LMCTFY (Google, Mountain View, CA, USA), and Apache Mesos (San Francisco, CA, USA). Compared to hypervisor-based virtualization, container technology allows for more efficient use of system resources by minimizing overhead and providing a lighter abstraction layer. Hypervisors operate full virtual machines with dedicated operating systems, whereas containers share the host OS, leading to faster performance and reduced resource consumption [94]. In contrast, container technology enables more efficient use of system resources, reducing the overhead.

To enhance flexibility and scalability in application management, container-based virtualization is often utilized in distributed and resilient edge clusters. For instance, Kubernetes (Mountain View, California, USA) and Apache Kafka (San Francisco, California, USA), both leveraging containerization, are implemented in such clusters [95]. Cloud environments favor containerization over hypervisor-based virtualization due to its reduced overhead and more efficient resource utilization. A similar security architecture is proposed in [96] to secure cloud environments using hypervisor-based virtualization. However, cloud providers are progressively adopting containerization to deliver faster and more scalable services by integrating virtualization technologies with self-service capabilities over network infrastructures.

In FC, virtual machines in cloud environments are insufficient to meet the requirements of fog nodes. Therefore, fog nodes use container technology to improve resource utilization and reduce service latency [97]. Docker is the most widely adopted container technology, directly managed by the main kernel [98]. Docker containers are configured through the command line interface (CLI) and Dockerfile, which contains all executable source code, libraries, and dependencies required for initialization.

ARM architectures—commonly used in mobile and low-power systems—are supported by container platforms like Docker and Kubernetes. Docker facilitates the creation of images specifically designed for ARM32 or ARM64 systems. Popular repositories such as Docker Hub provide optimized container images for these platforms. Table 4 illustrates the compatibility of container technologies with ARM-based edge hardware.

Table 4. ARM processor-based container technologies and edge hardware.

Ref. No	Container Technology	Edge Node Hardware	ARM Architecture
[91]	Docker	Raspberry Pi 3 Model B	Not specified
[99]	Docker, Containerd	Raspberry Pi 4 Model B and Raspberry Pi 3 Model B+	ARM v8
[100]	Docker	Raspberry Pi 3s	ARM Cortex-A53
[101]	KubeEdge, Kubernetes, Docker	ARM64 4-core CPU	ARM64
[102]	KubeEdge	Jetson AGX Xavier, Raspberry Pi 3	ARM64
[103]	Kubernetes	Raspberry Pi	Not specified
[104]	Docker	Raspberry Pi 4	ARM64
[105]	Container (Not specified)	Apple Mac Mini M1 2020 with Raspberry Pi and 16GB memory	ARM64
[106]	Docker	A Samsung Exynos 5250 SoC with 1.7 GHz Cortex A15 CPU and a non-virtualized host with 2 GB of memory	ARMv7
[107]	Kubernetes (K3S), Docker	Raspberry Pi 4 B+	ARM,x86
[108]	Kubernetes	Raspberry Pi	ARM

Finally, Kubernetes [109] is an open-source platform that manages the entire lifecycle of applications by organizing containers and hosting both VM- and container-based applications. Kubernetes optimizes resource usage by automating deployment, scaling, and maintenance, enabling rapid deployment of custom applications [110].

In summary, container-based virtualization technologies play a crucial role in improving the efficiency, scalability, and flexibility of EC and FC environments. By minimizing latency, optimizing resource usage, and supporting diverse hardware architectures, containers and orchestration tools such as Kubernetes have become indispensable in modern EC infrastructures. These advancements are essential in achieving real-time processing and service continuity, particularly in resource-constrained edge environments.

Container Isolation Technology

Containers enhance the efficiency of application management in EC environments. Containers improve application management in edge environments [111] by utilizing a single core and providing multiple isolated environments with lower overhead compared to VMs [112]. However, containers face limitations in security and isolation. To overcome these challenges, Unikernel, with its simplified architecture, offers higher security and stronger isolation [113]. Some EC systems adopt a hybrid virtualization mode that combines both VMs and containers to manage hardware resources and application services. Table 5 provides examples of virtualization technologies used in EC.

Table 5. Virtualization technologies in EC [113].

Platform	Virtualization Technique	Owners
EdgeX over Kubernetes [114,115]	Container	Linux (San Francisco, CA, USA)
FocusStack [116]	Container	Not specified
Akraino Edge Stack [117]	Container and Virtual Machine	Linux (San Francisco, CA, USA)
Azure IoT Edge [118]	Container	Microsoft (Redmond, WA, USA)
KubeEdge [119]	Container	Huawei (Shenzhen, China)
Cord [120]	Container and Virtual Machine	Open Network Foundation (Menlo Park, CA, USA)
AWS IoT Greengrass [121]	Container	Amazon (Seattle, WA, USA)
OpenEdge [122]	Container	Not specified
VideoEdge [123]	Container	Microsoft (Redmond, WA, USA)

Container-based EC systems have been increasingly adopted for various EC applications due to their ability to run efficiently on low-resource devices. For example, a container-based edge system was implemented on an ARM device in [124], where images were acquired through a container runtime/engine, processed locally, and subsequently transmitted to the cloud for storage and further analysis. This approach highlights the potential of containerized environments in optimizing local data processing and cloud integration.

Container runtimes such as Docker and Containerd are essential technologies for lightweight, efficient EC systems. In [125], two popular container runtime technologies, Docker and Containerd, were evaluated on Raspberry Pi 3 and 4 devices. Docker, owing to its lightweight nature, proved suitable for low-performance IoT devices, offering robustness, simplicity, and portability, while Containerd's open-source runtime further enhanced these advantages.

In addition to container runtimes, the performance of various ARM-based single-board processors has also been analyzed in the context of containerized applications. A study in [91] evaluated Docker containers on five ARM-based processors, including Raspberry Pi 2 Model B, Raspberry Pi 3 Model B, Odroid C1+, Odroid C2, and Odroid XU4. The findings emphasized the need to consider both task scheduling and containerization to optimize application execution efficiency of applications on edge servers. To address this, a joint task scheduling and containerization (JTSC) scheme was developed in [126]. Furthermore, ref. [127] highlights the importance of container virtualization technology in providing low-latency services in EC and proposes a two-stage optimization storage strategy to reduce

image file download time. This strategy optimizes file placement during both container initialization and runtime phases.

3.2. The Software and Tools of Edge Computing

3.2.1. Software Platforms

EC aims to reduce latency and enhance efficiency by processing data closer to the source. Software and tools in this domain play a crucial role in efficiently managing edge infrastructures, optimizing communication between devices, data processing, and resource utilization. There are several programming frameworks that have been specifically designed for EC, with popular choices including MXNet v1.9.1 (Apache Software Foundation, Forest Hill, MD, USA) [128], TensorFlow Lite v.2.17.0 (Google LLC, Mountain View, CA, USA) [129,130], CoreML v.2.3 (Apple Inc., Cupertino, CA, USA) [131], Caffe2 v.0.8.0 (Meta Platforms, Inc., Menlo Park, CA, USA) [132], PyTorch v.2.0 (Meta Platforms, Inc., Menlo Park, CA, USA) [133], and TensorRT v10.9.0.34 (NVIDIA Corporation, Santa Clara, CA, USA) [134]. TensorFlow Lite v.2.17.0 [129], developed by Google (Mountain View, CA, USA), is a version of TensorFlow v.2.15.0 optimized for mobile and edge devices. It uses advanced optimization techniques to perform deep learning model inference with a low latency on edge devices. In contrast, Caffe2 v0.8.0 [132], an updated version of the Caffe v.1.0 framework released by Facebook (Meta Platforms Inc., Menlo Park, CA, USA) in 2017, supports newer computational models such as mobile and distributed computing. It also offers cross-platform support, allowing developers to leverage GPU power on both cloud and edge devices.

FocusStack [135] is another key platform in EC, offering edge-cloud collaboration. By utilizing a container-based execution environment, FocusStack aligns edge devices with the capabilities of their infrastructure, ensuring scalability and efficiency for IoT applications [136]. This collaboration optimizes the deployment of IoT solutions by leveraging the strengths of both edge and cloud computing.

In [137], DeepThings, a framework for the adaptive distributed execution of CNN-based inference applications on IoT edge clusters, is proposed. This framework enhances the flexibility of EC systems by enabling dynamic adaptation to varying workloads, thereby improving computational efficiency in edge environments. Another important contribution is the use of Docker container-based analytics services at the edge for data processing, as highlighted in [138]. This approach reduces latency by processing data locally on edge devices before sending it to the cloud for further analysis. Moreover, the feasibility of deploying a deep learning framework for real-time video stream analysis on a Raspberry Pi is explored, with the MXNet v1.9.1 [139] framework selected for this task. Additionally, TensorFlow v.2.15.0 [140] employs a placement algorithm to distribute computational tasks based on estimated execution and communication times, optimizing the processing flow.

3.2.2. Communication Protocols

The computational workflow is distributed across the IoT devices, edge nodes, and cloud data centers depending on the application's functional requirements and QoS parameters. In this process, data are transmitted from the IoT device to the edge and then from the edge to the cloud using different communication protocols. Lightweight protocols such as the Constrained Application Protocol (CoAP) and Bluetooth Low Energy (BLE) are commonly employed to connect resource-constrained end devices to edge gateways [141].

To facilitate data flow and monitoring on the edge server, foundational software tools such as Eclipse Mosquitto (implementing the MQTT protocol), Node-RED (a flow-based programming environment), and SQLite (a lightweight SQL database engine) are commonly deployed [142]. MQTT, recognized for its minimal bandwidth and power consumption,

follows a lightweight publish/subscribe (pub/sub) model and is widely adopted in edge computing scenarios. Eclipse Mosquitto was selected due to its open-source nature and compatibility with constrained devices. Figure 5 illustrates the operational interactions among these components within the edge environment.

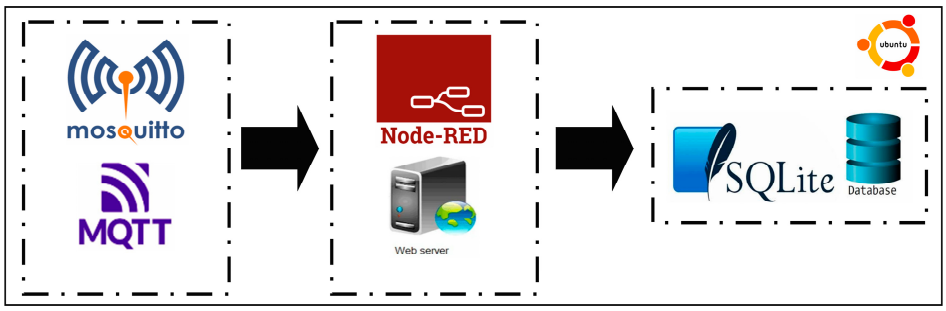


Figure 5. Software infrastructure and services deployed on the edge server [142].

MQTT, as a pub/sub protocol, is supported by various brokers including Eclipse Mosquitto and HiveMQ. To enhance security in MQTT communications, the MQTLS protocol proposed in [143] introduces a mechanism that ensures that brokers read messages only when necessary, enabling selective message delivery between publishers and subscribers. Currently, MQTT security is primarily ensured through the use of the TLS protocol between the client and broker, as illustrated in Figure 6.

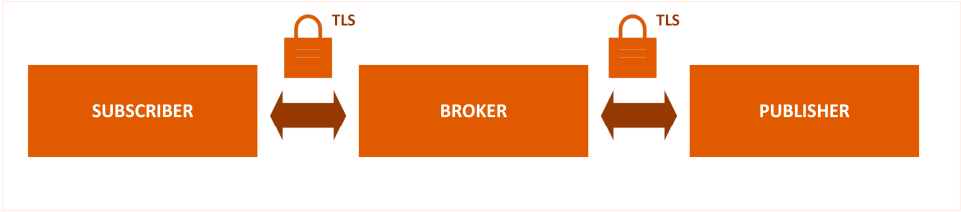


Figure 6. The security model in the pub-sub model in TLS [143].

AMQP is an open-standard, TCP-based application layer protocol designed for message-oriented environments [144]. Similarly, the Extensible Messaging and Presence Protocol (XMPP) is utilized in instant messaging applications and supports both synchronous (request-response) and asynchronous (pub-sub) communication models. XMPP also relies on TCP as its underlying transport protocol [145].

The MQTT architecture consists of publishers (message-generating entities), intermediaries (components that ensure QoS and security), and subscribers (message receivers) [146]. Table 6 presents a comparative overview of commonly used application-layer communication protocols, emphasizing their respective strengths and use cases in edge and IoT networks.

Table 6. Comparison of common application protocols [146].

Application Protocol	Publish/ Subscribe	Request/ Response	Transport	Security	Quality of Service(QoS)
Constrained Application Protocol (COAP)	Yes	Yes	UDP (User Datagram Protocol)	DTLS (Datagram Transport Layer Security)	Yes
Minimum Message Queuing Telemetry Transport (MQTT)	Yes	No	TCP (Transmission Control Protocol)	SSL (Secure Sockets Layer)	Yes
Advanced Message Queuing Protocol (AMQP)	Yes	No	TCP	SSL	Yes
Extensible Messaging and Presence Protocol (XMPP)	Yes	Yes	TCP	SSL	No

3.3. The Hardware and Tools of Edge Computing

Edge devices include smartphones, laptops, desktop computers, and distributed servers (e.g., in base stations) [147]. A summary of some end devices is provided in Table 7. Additionally, Table 6 presents hardware that supports the execution of AI algorithms for edge devices, edge nodes, and edge cloud systems in AI of Things (AIoT) applications [16,148].

Table 7. Some edge devices and owners [16,148].

Productions	Owners
TPU	Google
DianNao family	Cambrain
Turing GPUs	NVIDIA Corporation
7 Series FPGA	Xilinx
HiSilicon Ascend Series	Huawei
Exynos 9820	Samsung
Xeon D-2100	Intel
TrueNorth	IBM

4. The Application of Edge Computing System

Power systems, encompassing a wide range of components from energy transmission to distribution, are becoming increasingly intelligent with the integration of EC technologies. The increasing need for data processing and the growing complexity of traditional energy distribution networks have made these systems highly dependent on the capabilities of EC. This technology offers innovative solutions in various applications, including data transmission, monitoring and control, microgrid systems, smart meters, anomaly detection, state estimation, energy management, and fault location. This section will explore the applications of EC in power systems, emphasizing its potential advantages and its integration into modern energy grids.

4.1. Data Transmission

Data transmission in power systems plays a critical role in the efficient management and monitoring of energy networks. In these systems, large volumes of data are collected from sensors, smart meters, and other monitoring devices to facilitate the oversight of energy generation, transmission, and distribution. Figure 7 illustrates the EC architecture for power transmission applications. To ensure secure and timely data transfer, specialized communication protocols and infrastructures are required to ensure secure and timely data transmission [149]. A three-tier architecture has been proposed for implementing EC in smart grids, formed by the cloud-edge-thing continuum [150]. The total response time is determined by the sum of transmission and processing delays. In cases where IoT devices generate massive amounts of data but have limited computational resources, high network latency becomes unacceptable as they fail to meet QoS requirements. Unlike traditional cloud services, EC addresses the needs of IoT applications more efficiently by enabling real-time data collection and analysis with low latency [151]. Additionally, in [152], a fine-grained adaptive power control scheme is introduced to reduce latency, which optimizes data transmission paths using adaptive power management techniques.

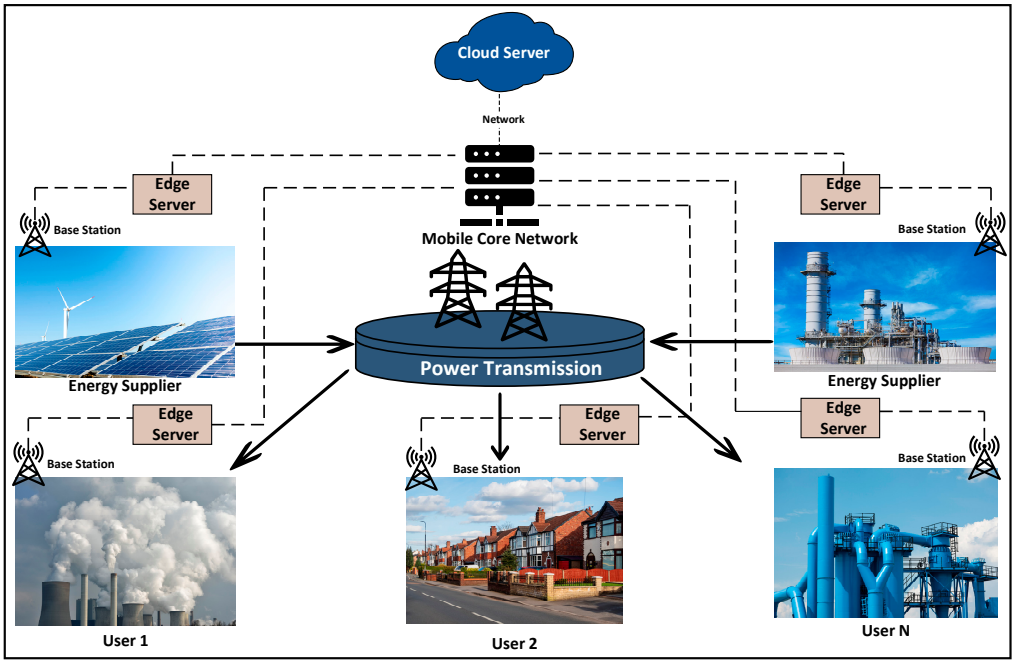


Figure 7. EC architecture for power transmission applications.

Various optimization strategies have been investigated to enhance power transmission and energy efficiency. In [153], a long-term stochastic optimization problem was decomposed into multiple subproblems to maximize the transmission success rate. This approach integrates power control coefficients and aggregation coefficients, aiming to optimize the minimum time-average transmission success rate (TATSR) by determining the optimal coefficients for each period while accounting for the dynamic statistics of randomly arriving data. Additionally, power allocation strategies designed to improve energy efficiency were explored in [154]. This study employs fractional programming and convex optimization techniques, demonstrating that secondary users can achieve substantial energy savings even under average transmission power constraints.

Furthermore, EC solutions for power transmission have been explored in [155], emphasizing the role of IoT sensor technologies in preventing faults in transmission lines. These technologies not only enhance transmission power capacity but also reduce economic losses by improving reliability. In the domain of voltage/VAR control (VVC) for distributed energy resources (DER), ref. [156] proposed an artificial neural network (ANN)-based strategy, enabling DER inverters to operate autonomously and overcome the limitations of conventional control methods. In a related effort, ref. [157] introduced a hierarchical-hybrid architecture for VVC in grids with high penetration of distributed generators (DG). Similarly, in [158], a three-tier voltage control framework tailored for grid clusters equipped with smart PV-based inverters (SPVIs) was presented, aiming to maintain voltage stability and minimize power losses. Finally, ref. [159] developed a monitoring system that combines Zigbee wireless communication with EC for remote transmission line monitoring. This system demonstrated low energy consumption, strong anti-interference capabilities, and improved fault detection accuracy. The proposed EC architecture for power distribution applications is illustrated in Figure 8.

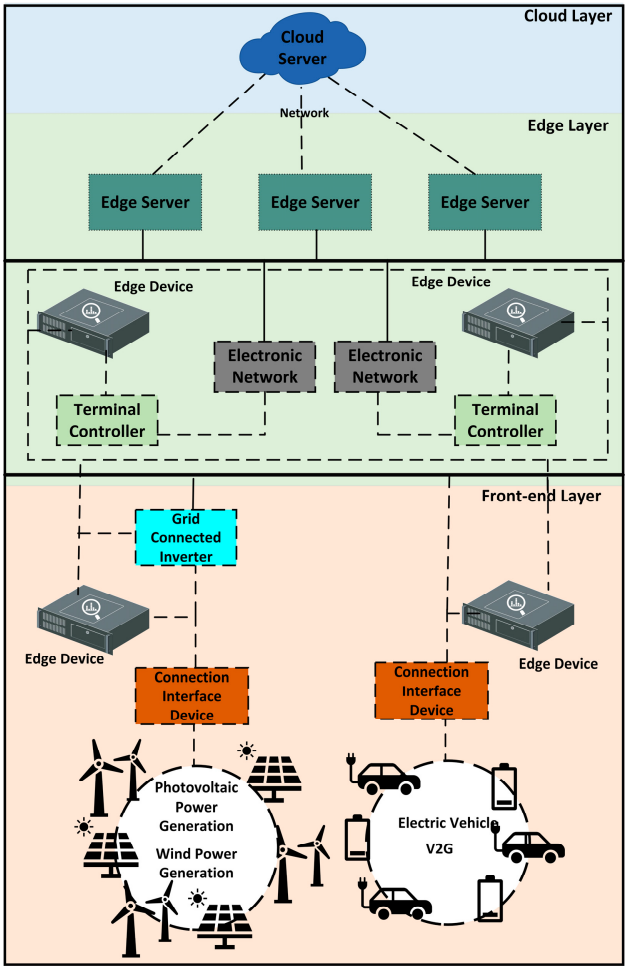


Figure 8. Proposed EC architecture for power distribution applications.

4.2. Monitoring and Control

In power grids, the continuous monitoring and transmission of critical data in distribution networks enhance operational efficiency. However, the increasing density and volume of exchanged information across the network can degrade QoS, particularly under centralized processing models. To address this challenge, numerous studies have investigated decentralized computing approaches and the integration of EC into power distribution systems.

Several studies have focused on integrating IoT paradigms into the Internet of Services (IoS) to enable decentralized computation and management. For instance, ref. [160] investigates decentralized computing potentials by embedding IoT principles into IoS architectures, while [161] develops an online monitoring system designed for transient electromechanical simulation in distribution grids. Similarly, ref. [162] proposes an EC-based monitoring framework for evaluating reactive power behavior in smart distribution transformers, aiming to improve real-time data analytics and responsiveness.

The potential of cloud-edge collaborative computing models has also been extensively explored in intelligent monitoring applications. Ref. [163] introduces a hybrid architecture to enable real-time diagnostics of power distribution equipment via a cloud-edge framework. Likewise, ref. [164] improves grid security and reliability by integrating IoT and EC into a smart grid monitoring platform. Collectively, these contributions highlight how EC enhances operational performance and system resilience in power distribution networks.

Another critical dimension of EC applications involves resource management and service orchestration in complex power systems. In [165], an EC-based service scheduling

approach is proposed to address the increasing computational and bandwidth demands in smart city and Power Distribution IoT (PD-IoT) infrastructures. Complementarily, ref. [166] introduces a task allocation strategy for real-time monitoring and early warning services, aiming to optimize the utilization of computational resources.

In addition to monitoring and resource allocation, EC has been employed to enhance system integration and metering infrastructure. In ref. [167], a smart integrated terminal is designed using EC principles to improve metering efficiency and operational coordination. Similarly, ref. [168] introduces an EC-based power grid mapping structure for the digital management of distributed power networks. To establish a structured framework for power distribution applications, ref. [169] defines a four-layer EC-based architecture, comprising cloud, network, edge, and terminal layers, which serves as a reference model for future implementations, as shown in Figure 9. Further expanding on this architecture, ref. [170] proposes a "cloud-side-edge collaboration" framework to enhance the scalability and availability of substation operation support systems. This model ensures seamless coordination between monitoring, data processing, and decision-making tasks across multiple layers.

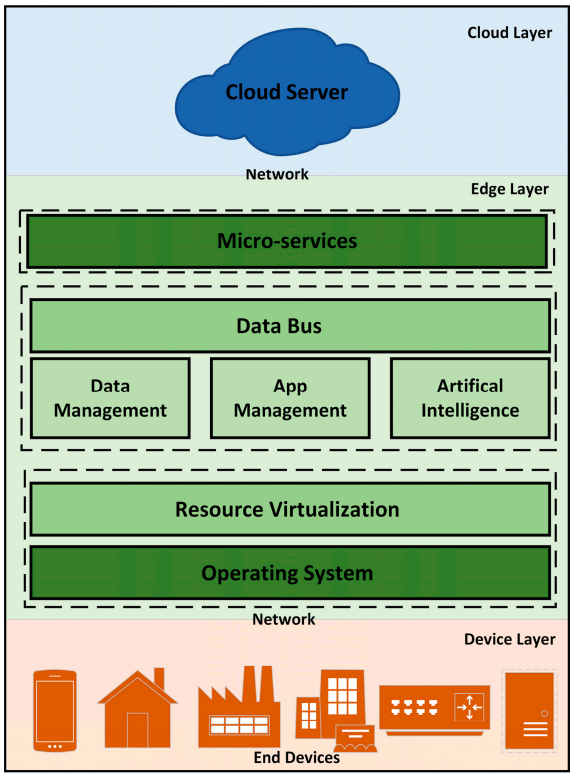


Figure 9. The architectural framework of the power distribution network based on EC [169].

Moreover, EC enables optimization of communication latency and computational load management. For instance, ref. [171] develops a node scheduling model based on EC that employs a genetic algorithm with a fighter search strategy to reduce transmission delays and minimize economic costs.

Despite these advancements, traditional centralized monitoring systems remain insufficient to address the growing complexity and dynamism of modern power grids. To overcome these limitations, ref. [43] proposes a five-tier EC framework for real-time monitoring. As shown in Figure 10, the system automates line monitoring by processing image data from IoT devices at the edge, thereby enhancing threat detection and response times.

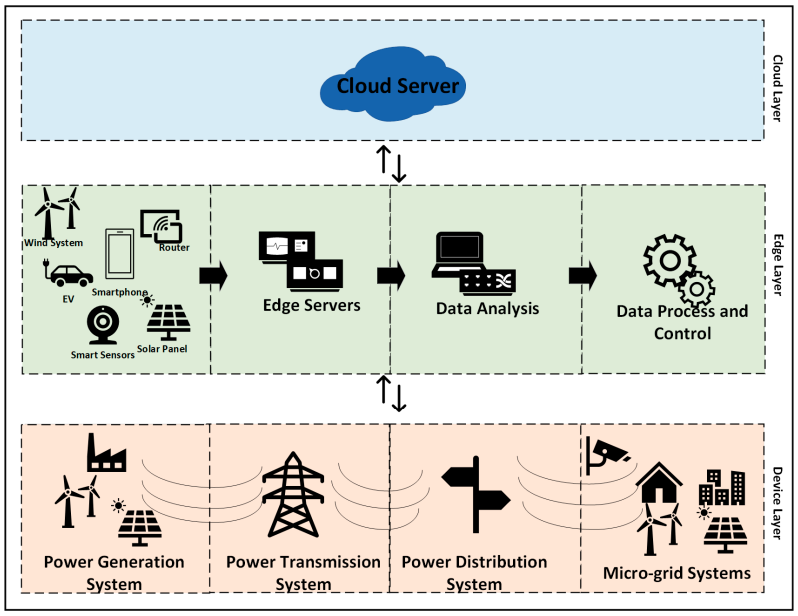


Figure 10. Design principles and functional structure of EC architecture in smart grid applications.

4.3. Microgrid System

EC architecture in microgrid systems represents an innovative approach that enhances energy management and data processing. This architecture enables the efficient processing, storage, and management of data from energy equipment, thereby enhancing real-time decision-making and operational efficiency. As shown in Figure 11, the EC architecture for microgrids follows a three-tier structure comprising device, edge, and cloud layers. The device layer includes of microgrid energy equipment, the edge layer incorporates edge platforms that provide computing, storage, and application functionalities, while the cloud layer hosts various cloud-based services.

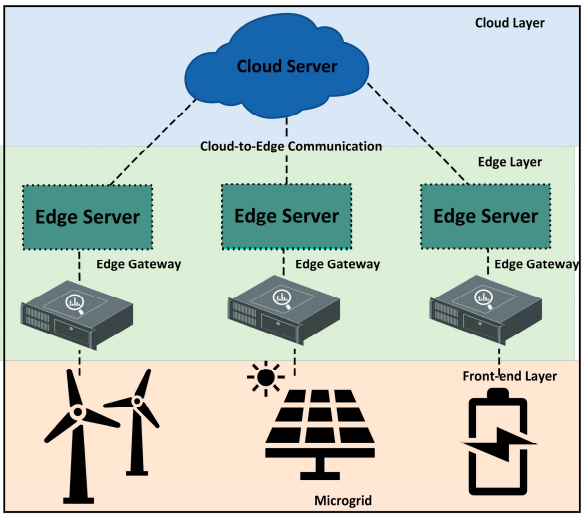


Figure 11. Architectural overview of EC integration in microgrid systems.

In microgrid applications, the edge layer plays a crucial role in data collection, transmission, and device control by utilizing various network communication protocols such as MQTT and CoAP, as described in Edge Computing Software Protocols. Each edge gateway device collects real-time operational data from energy equipment and transmits it to the edge server, enabling local data processing and efficient network management. This decentralized approach alleviates the processing burden on the cloud while enhancing

system responsiveness. The cloud server subsequently analyzes the aggregated data to support decision-making processes. For instance, Figure 12 illustrates a microgrid system implemented in a rural community in central China, where EC-based energy management techniques enable real-time forecasting of renewable energy generation and load demand [171]. Equipment scheduling microservices further optimize energy usage by managing generator outputs and battery charge–discharge cycles in real-time.

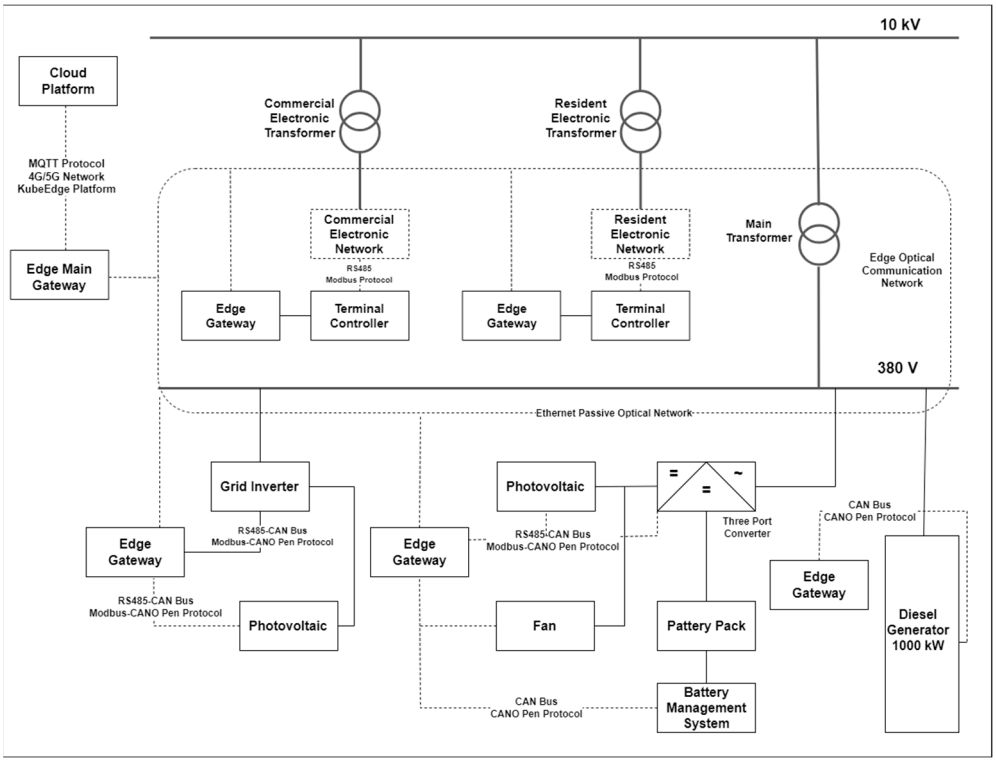


Figure 12. Edge computing-based microgrid architecture in central China [171].

Several studies have explored the integration of EC into microgrid energy management. Ref. [172] investigates energy supply scheduling in multi-access edge computing (MEC)-enabled microgrids, proposing an optimization framework that considers uncertainties and delay constraints in energy production and consumption. Addressing similar challenges, refs. [173,174] introduce a risk-sensitive energy profiling approach for MEC networks, utilizing multi-agent deep reinforcement learning (MADRL) to enhance decision-making under dynamic conditions. Additionally, ref. [175] presents a power control framework that integrates EC with reinforcement learning, enabling real-time network state detection and power flow optimization through edge intelligence. Figure 13 illustrates the system model designed in [172,173] general terms.

The use of machine learning (ML) in EC-based microgrid management has gained significant attention. Ref. [176] proposes an ML-driven EC architecture for energy management, utilizing Kafka Streams and open-source IoT platforms to predict energy consumption and production. In this model, edge devices collect data from individual prosumers and apply AI algorithms for real-time energy forecasting. Similarly, ref. [177] examines a microgrid operating in islanding mode, focusing on dynamic response mechanisms to maintain voltage and frequency stability. Further, ref. [178] explores power supply management, power balance, and malicious behavior detection in IoT-enabled smart grids, introducing EC-based prediction strategies and task rating-based privacy protection mechanisms. To enhance microgrid efficiency, ref. [179] presents a hybrid EC framework that leverages short-term electricity generation and consumption forecasting, incorporating

wind power and smart meter data. In addition, ref. [180] integrates AI-driven algorithms into MEC-based microgrid optimization, proposing a neural network-based identification method that employs online adaptive dynamic programming to eliminate dependency on predefined system models. Finally, ref. [181] highlights the significance of EC connectivity with control centers in microgrids, emphasizing its role in addressing real-time electricity demand scenarios.

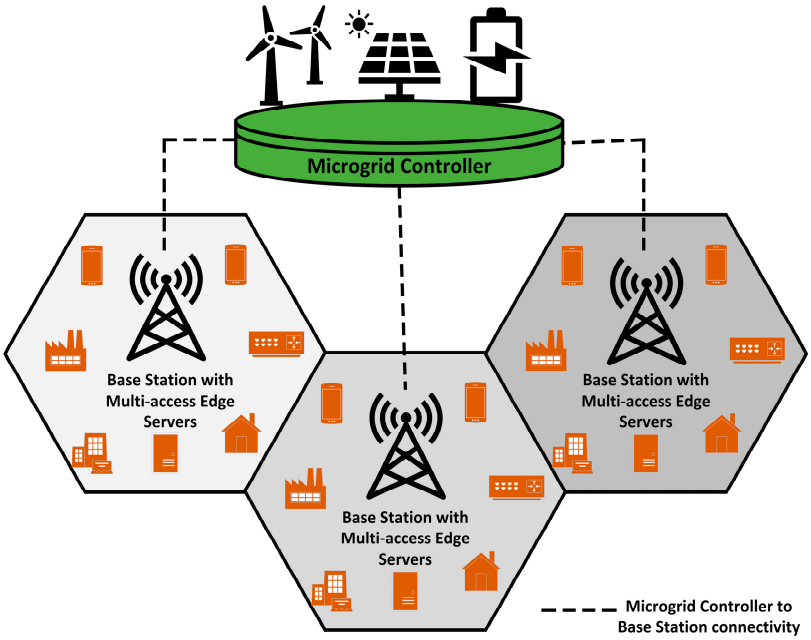


Figure 13. EC-based system model for microgrid applications [172,173].

4.4. Smart Metering System

Advanced metering infrastructure (AMI) is a critical component of modern power systems, enabling bidirectional data communication between smart meters and utility centers [182]. The integration of EC and edge intelligence with AMI enhances operational efficiency by reducing communication costs, protecting user privacy, and increasing situational awareness [183]. Figure 14 presents an overview of the EC-enabled smart meter architecture. In such systems, distributed deep learning based on federated learning (FL) has proven to be an effective approach for applications like non-intrusive load monitoring (NILM).

Smart meters, as key consumer-side components of AMI, perform several essential functions, including real-time pricing updates, energy consumption monitoring, power loss detection, power quality assessment, P2P communication with other meters, and remote switch control [184]. The meter data management system (MDMS) plays a crucial role in processing and storing the large volumes of data collected from smart meters via data concentrators. Furthermore, MDMS supports decision-making processes that optimize energy consumption by integrating with other application systems. Smart meters collect real-time energy consumption data from IoT devices within the home area network (HAN) and transmit these data to the MDMS, where control instructions are generated. To enhance communication efficiency, ref. [185] proposes an aggregation-based Low Power Loss Area Network (RPL) scheme for power distribution networks. This scheme modifies the ContikiRPL protocol using the Low Power Personal Area Network (6LoWPAN) protocol, improving communication between smart meters.

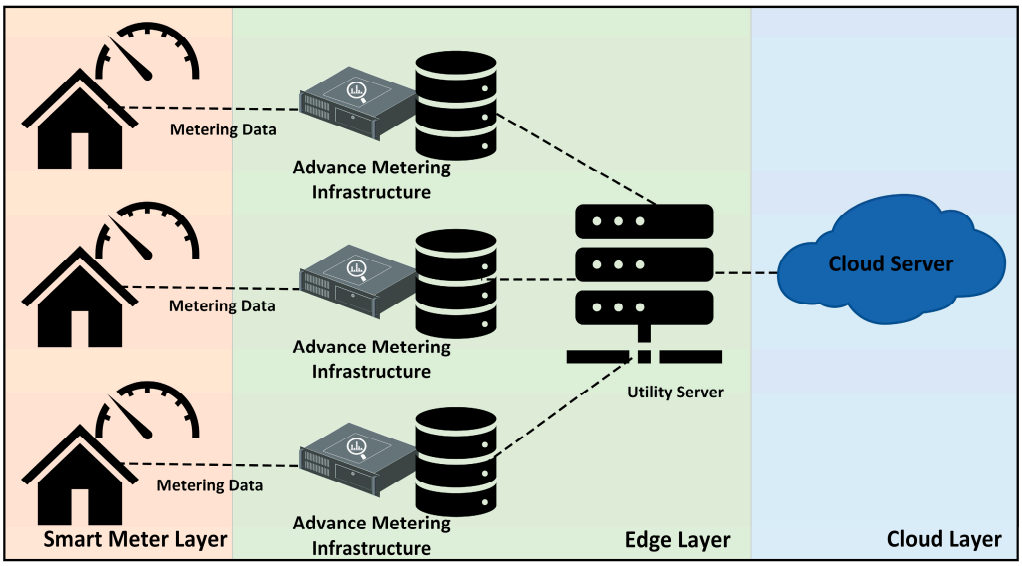


Figure 14. System design and interaction between EC and Smart Meters.

EC enables distributed computing capabilities, supporting energy control and enhancing the functionality of smart grids (SGs) through deep learning models [186,187]. In [188], an optimization framework is proposed for code and data placement to enhance system stability and cost-effectiveness. Moreover, an advanced AMI model presented in [178] facilitates accurate electricity price forecasting, local-level service analysis, resource allocation optimization, and cost reduction by improving metering efficiency. Similarly, ref. [189] introduces a knowledge-based deployment strategy to optimize the energy consumption and lifespan of smart meters. The role of smart metering in IoT-based smart grids extends beyond conventional applications to improving power quality and reliability. Ref. [190] examines the applicability of smart metering technologies in modern power grids, emphasizing their role in enhancing grid modernization efforts. Furthermore, ref. [191] introduces a LoRa-based smart metering system designed to minimize energy consumption during data transmission, thereby enhancing the overall efficiency of the network. This system employs deep learning-based compression and optimization techniques, enabling energy-efficient data transfer with reduced latency. Ref. [192] develops an EC-supported data analytics system that utilizes a multi-process architecture for smart meters. The results demonstrate that incorporating edge analytics into smart meters significantly enhances data processing performance, with multiprocessing further improving system responsiveness.

4.5. Anomaly Detection

Various anomaly detection approaches have been discussed in the literature. For instance, the method proposed in [193] for smart meter fault detection is a decision tree (DT) approach that filters abnormal data using meter data analytics and classifies them according to energy loss levels. This method provides online detection and verification for large-volume meters, thereby reducing the need for on-site inspections. To address the latency issues in data transmission, ref. [194] proposes anomaly detection using prediction algorithms such as Deep Neural Network (DNN), Support Vector Regression (SVR), and K-Nearest Neighbor (KNN). The study shows that DNN is the most effective solution with low latency (1.25 ms). The system ensures seamless data flow and collaboration between Smart Meters, Concentrators, Edge-MDMS, and Cloud-MDMS, enabling early detection and alerting of anomalies.

In another approach, ref. [195] developed a hybrid model to identify anomalies in building energy data. This model combines polynomial regression with Gaussian distribu-

tion to detect anomalies in school electricity consumption data. Focusing on cybersecurity and data security, ref. [196] proposes an EC-based anomaly detection method for AMI. This method is implemented at the endpoint of the data concentrator, effectively mitigating the risk of privacy leaks and reducing both communication latency and computation time. Successful results were achieved with 99.05% detection accuracy and a 0.74% false detection rate. Similarly, ref. [197] introduces privacy-preserving anomaly detection methods through FL, an innovative approach that leverages distributed edge devices to detect anomalies while maintaining data privacy. By enabling local data processing, FL eliminates the need to transfer sensitive information to centralized servers, ensuring user privacy. This is particularly beneficial in scenarios like transformer-based models for sequential anomaly detection, where decentralized training enhances model accuracy and reduces communication overhead.

In the context of Edge-AI, ref. [198] highlights the convergence of EC and AI, which enables the deployment of advanced AI models to be deployed closer to users, significantly enhancing performance. A key framework in this domain is Federated Continual Learning (FCL), which supports distributed learning across clients while preserving data privacy. FCL ensures stable model performance in dynamic environments by retaining knowledge from previous tasks, making it ideal for anomaly detection applications that require real-time adaptability across distributed edge devices.

Finally, ref. [199] developed a Granger causality test-based approach to detect electricity theft among low-voltage (LV) users. This method detects electricity theft with data measured at the distribution transformer unit (DTU) and avoids false alarms caused by corrupted data.

4.6. State Estimation

In modern energy systems, SE plays a crucial role in monitoring power grids by analyzing erroneous measurements and determining the correct values of unknown variables. With the increasing demand for real-time control, SE requires a high bandwidth and powerful computing resources. To address this challenge, solutions such as performing SE at distributed edge nodes and transmitting the results to the control center have been proposed. In this context, ref. [200] presents a Cloud-IoT-based architecture solution that combines the benefits of CC and EC for condition estimation in smart grids. This solution enhances operational flexibility and adaptability by separating measurement data from physical devices using virtualization technologies. Similarly, ref. [201] proposes a multi-layered hierarchical framework that integrates cybersecurity measures, performing condition estimation on edge devices while utilizing security mechanisms to detect malicious devices using Phasor Measurement Unit (PMU) data.

To mitigate false data injection attacks, ref. [202] developed an edge-based FL framework. This approach addresses challenges such as distributed datasets and unknown system parameters while improving detection accuracy through a mechanism that incentivizes data owners. Ref. [203] examines how false data injection (FDI) attacks can affect the SE system. Furthermore, ref. [204] introduces a data-driven switching state identification algorithm for renewable energy integration and frequent line switch states. The EC framework distributes computational tasks to smart terminals, thereby reducing communication and computational load and enhancing algorithm performance. Additionally, ref. [205] presents an NSGA-III-based optimization, leveraging EC to address network partitioning and edge server placement challenges. This approach optimizes network performance by minimizing edge server costs.

4.7. Energy Management System

Studies on EMS primarily focus on objectives such as optimizing energy consumption, enhancing demand response, and improving user satisfaction. Ref. [206] examines the impact of IoT and EC technologies on energy management in smart cities, proposing an approach that enables instant energy control and long-term savings by processing data collected from IoT devices using deep learning techniques. Ref. [207] developed a customizable home energy management system (HEMS) middleware platform that is based on microservice architecture and low-cost hardware. This system, utilizing an edge computing approach, minimizes cloud dependency by performing energy management closer to the user while also achieving high performance with low latency and efficient resource utilization.

In the context of EMS, various strategies have been proposed to optimize both energy supply and demand, as well as improve efficiency and user satisfaction. For example, ref. [208] focuses on optimizing both supply-side energy allocation and demand-side load response in smart grids using privacy-preserving distributed algorithms. This approach aims to balance energy distribution across the grid while safeguarding users' privacy. Ref. [209] presents an EC architecture that integrates unified learning and deep reinforcement learning (DRL) algorithms to optimize energy consumption in building energy management systems, aiming to improve energy efficiency.

To address energy cost reduction in residential settings, ref. [210] developed an IoT edge-based energy management system and proposed a middleware architecture for HEMS. This system adopts a load-shifting and reinforcement-learning approach, performing planning based on energy production, consumption, and device runtime. Ref. [211] proposes a three-tier edge-cloud collaboration-based residential energy management (ECCREM) architecture to reduce energy costs and stabilize demand fluctuations. Using Stackelberg and Lyapunov-based pricing algorithms, this architecture optimizes energy demand and aims to increase user satisfaction by up to 20% with priority-based scheduling strategies.

4.8. Fault Location

A system developed using EC stores transition amplitude ratios at edge nodes and utilizes a radial basis function (RBF) neural network to accurately detect faults in networks with hybrid feeders [174]. Transients are sampled at measurement nodes and transmitted to the edge nodes, which identify the faulty branch and perform the detection process. To address overload issues caused by centralized management in large-scale digital distribution networks, ref. [175] proposes a power grid mapping-based EC structure that improves network efficiency by optimizing the use of centralized management resources. In [212], an innovative line selection and fault detection method for distribution networks with cloud-edge-terminal hierarchical architecture is proposed. When a fault occurs, the traveling wavefront arrival times are transmitted via optical fiber to the edge computing gateway. Here, large volumes of fault data are filtered, and invalid and noisy information is eliminated. The time characteristic matrices generated from the valid data are transferred to the cloud layer, where line selection is performed. Then, fault localization is handed over to the edge layer, and fault localization is performed. The result is transmitted to the terminal layer to trigger the relevant circuit breaker. Finally, the correct fault information is transmitted to the cloud and presented to the user in a pop-up window format. Similarly, ref. [169] proposes an EC-based fault detection method to promptly identify faults in distribution networks and improve electricity supply reliability. The EC-supported fault monitoring and control framework is depicted in Figure 15. In [213–216], a hierarchical fault monitoring system for large and complex distribution networks is developed, offering low latency and localized data processing with EC. This system aims to overcome the

limitations of traditional methods, particularly in detecting and processing small current faults, such as single-phase ground (SLG) faults.

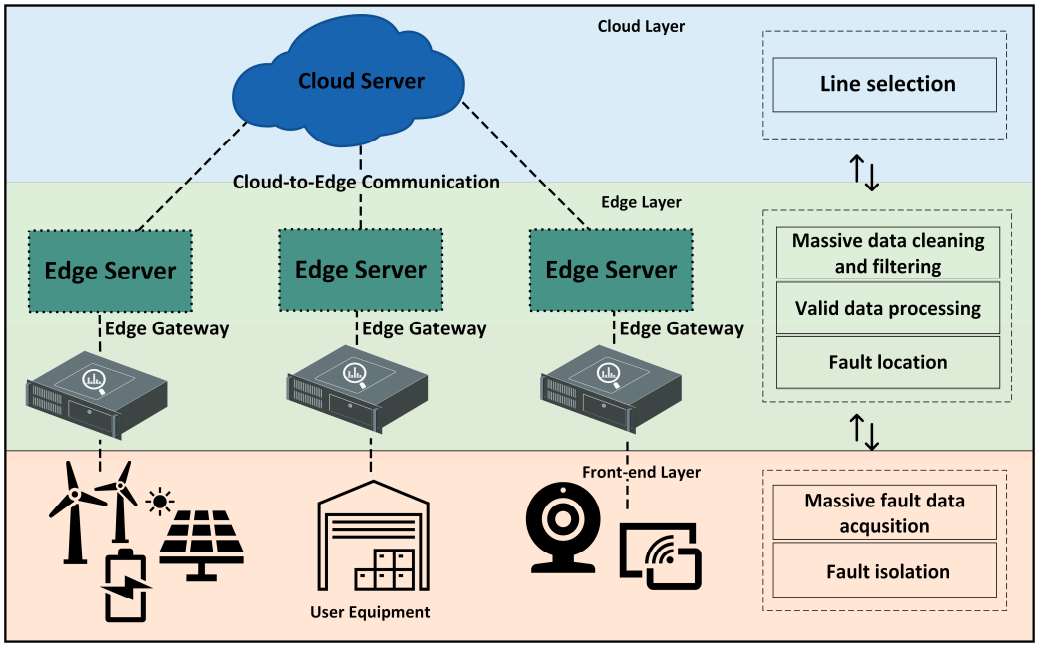


Figure 15. Proposed EC-supported fault monitoring and control framework [212].

Table 8 summarizes the power system application of EC.

Table 8. The survey on the integration of edge computing in applications.

Ref. No	Scenarios	Key Technologies
[152]	Data Transmission	Centralized Software Defined Network (SDN) and EC
[153]	Data Transmission	TATSR
[154]	Data Transmission	Optimal Power Allocation Strategy with Transmission Power Constraints
[155]	Data Transmission	Information/Digital Technologies and Artificial Intelligence Planning Techniques
[156]	Data Transmission	ANN and VVC and EC
[157]	Data Transmission	VVC and Reactive Power Control
[158]	Data Transmission	Voltage Stability
[159]	Data Transmission	Zigbee Wireless Communication and EC
[161]	Monitoring and Control	Transient Electromechanical Simulation
[162]	Monitoring and Control	Reactive Power Condition Online Monitoring and EC
[163]	Monitoring and Control	A priori Frequent Item Set Algorithm and EC
[165]	Monitoring and Control	PD-IoT System
[217]	Monitoring and Control	A GA Based on Predator Search Strategy
[172]	Microgrid System	Model-Based Deep Reinforcement Learning (MDRL) and EC
[173]	Microgrid System	MADRL
[174]	Microgrid System, Fault Location	Radial Basis Neural Network Function
[175]	Microgrid System, Fault Location	DRL and EC
[176]	Microgrid System	ML Algorithm and EC
[178]	Microgrid System, Smart Metering System	Hierarchical Decision-Making Strategy Based on Prediction Strategy and Task Grading (HDTG) and Real-Time Electricity Price Forecast
[179]	Microgrid System	Edge-AI Algorithm
[180]	Microgrid System	Neural-Network-Based Identification Scheme
[183]	Smart Metering System	NILM
[185]	Smart Metering System	6LoWPAN Protocol and EC

Table 8. *Cont.*

Ref. No	Scenarios	Key Technologies
[187]	Smart Metering System	DL and EC
[189]	Smart Metering System	Knowledge-based Usage Strategy for Smart Meters
[191]	Smart Metering System	EC in Long Range (LoRa) and DL Based Compression–Decompression Model
[193]	Anomaly Detection	DT to Filter the Abnormal Data
[195]	Anomaly Detection	Polynomial Regression and Gaussian Distribution
[196]	Anomaly Detection	KDDCUP99 Datasets
[202]	State Estimation	FL Framework for False Data Injection
[204]	State Estimation	Data-Driven Algorithm
[205]	State Estimation	Genetic Algorithm III (NSGA-III)
[208]	Energy Management System	Privacy-Preserving Average Consensus Algorithm
[209]	Energy Management System	FL and DRL Algorithms
[210]	Energy Management System	Reinforcement Learning Algorithm
[211]	Energy Management System	Stackelberg and Lyapunov Algorithms
[212]	Fault Location	TWAM
[218]	Fault Location	Power Signal Fault Signal Analysis and EC

5. The Security and Privacy of Edge Computing

EC has become a critical technology for enhancing privacy, security, and resilience against cyber-attacks in smart grids. Additionally, it addresses key challenges such as the efficient deployment of IoT devices and the management of processing loads. Several studies have made significant contributions by focusing on security and privacy requirements. Ref. [219] examines the cybersecurity challenges and solutions for cyber-physical systems (CPS) within the IIoT–edge computing context, highlighting common threats such as DoS and ransomware. It also discusses various security technologies, including machine learning, blockchain, and encryption, as essential tools for improving security and real-time data protection in IIoT–edge environments, while outlining future research directions in this area. Refs. [50,220] analyzed cybersecurity vulnerabilities in smart grids, including power outages, protocol failures, and breaches, and proposed solutions to enhance resilience using secure communication and intrusion detection technologies. Additionally, security and privacy concerns in EC-supported IoT systems have been thoroughly explored [50], with an analysis of threat vectors and vulnerabilities by ETSI standards. Solutions against these threats and future roadmaps have also been proposed [221]. Infrastructure, security, privacy, and scalability challenges are discussed in detail, examining the integration of edge computing and IoT technologies in smart cities through applications such as energy management, public safety, healthcare, and transportation [222]. Ref. [223] evaluates decentralized edge architectures, edge nodes, and gateways, examining their role in various Industry 5.0 applications such as smart cities, autonomous systems, and healthcare. EC introduces security and privacy challenges alongside deploying heterogeneous devices and the complexities of big data processing. Ref. [224] addressed security and privacy issues in EC architecture, discussing the role of ML and DL algorithms in combating these threats, especially to reduce the risks associated with a large attack surface. For IoT systems, ref. [225] proposed FC and EC approaches to facilitate the handling of confidential user data and improve data security. This study proposes enhancing security by continuously monitoring system resources and restricting data transmission to processed information. Additionally, a System Security Manager is introduced to mitigate vulnerabilities associated with device-level attacks.

As these advancements in security and privacy through EC are crucial, they enable the development of more resilient and secure Digital Twins (DT) for grid systems, especially

in predictive maintenance and resilience applications [226]. This makes it an essential component for developing DT in grid resilience and predictive maintenance systems. In line with this, ref. [227] proposes an FL-based DT framework called SAINT, designed to address challenges in resource scheduling for 5G edge computing-empowered distribution grids. SAINT optimizes latency, accuracy, and security by reducing iteration delay, enhancing model recognition, and supporting intelligent resource scheduling while considering energy consumption and access priority.

Several studies have focused on how blockchain technology can reduce the processing load on the cloud and minimize the latency between IoT devices and the cloud. The integration of blockchain and EC provides a decentralized approach to address security and data integrity issues in IoT networks, providing a reliable authentication and data storage mechanism [228,229]. Although EC improves the security of sensitive data, it poses security challenges such as authentication of IoT devices and edge servers and risks such as vulnerability to malicious attacks [230]. Accordingly, the hybrid centralized and blockchain-based authentication architecture proposed by [231] provides an efficient and decentralized authentication mechanism for IoT systems using edge computing. While deploying edge servers for centralized authentication, it uses a lightweight blockchain network to increase security, reduce transaction costs, and meet the scalability and real-time performance needs of IoT devices. Blockchain technology strengthens the security of EC with its immutable, hack-resistant, and decentralized data storage features, securing transactions between IoT devices and acting as a trusted database [232]. In addition, blockchain's consensus mechanisms provide an innovative method for distributed database management and new payment models for network services in edge environments.

The integration of EC and blockchain combines CC, data processing, and access control in a secure and unified platform, enabling rapid deployment of computational services to edge servers and accelerating the development of the Industrial Internet of Things (IIoT). In this context, a blockchain-enabled edge intelligence framework is proposed to provide secure and flexible edge service management, including cross-domain sharing-based scheduling of edge resources and a credit-based edge transaction approval mechanism. Numerical results show that these approaches provide significant improvements in both edge service cost and capacity [233]. Furthermore, a lightweight secret key agreement protocol based on public key with self-authentication has been developed to enable secure communication in resource-constrained edge devices. These mechanisms work together to provide IIoT security guarantees such as authentication, auditability, and confidentiality [234].

In addition to improving security, deep learning-based user authentication, combined with hybrid encryption, enhances blockchain-aided data storage while optimizing task offloading in MEC. This approach improves data integrity, boosts computational efficiency, and further strengthens security in decentralized environments [235]. The concept of "decentralizing edge devices" leverages blockchain technology to establish a distributed ecosystem where edge devices operate independently without reliance on a central authority. Blockchain ensures secure interactions by tracking, verifying, and regulating the behavior of these devices while managing their data exchange. This decentralized structure enables edge devices to support distributed computing platforms, facilitating collaborative execution of computational tasks without a central governing entity [236]. Moreover, blockchain-based decentralization can lead to more adaptable and secure distributed networks, fostering the development of decentralized applications. In this context, a power allocation platform has been proposed, utilizing a decentralized edge computing model to enhance resource efficiency and security [237].

While deep learning-based decentralized authentication and hybrid encryption significantly enhance security and efficiency, their implementation presents notable challenges. One of the key concerns is the computational overhead introduced by these mechanisms, particularly in resource-constrained edge devices. Processing cryptographic operations and consensus mechanisms may lead to increased energy consumption and latency, affecting real-time performance. Additionally, integrating decentralized authentication with existing legacy IoT systems remains a challenge due to differences in protocol compatibility and infrastructure requirements. Addressing these challenges requires optimizing cryptographic algorithms for lightweight execution and developing interoperability frameworks to ensure seamless integration with traditional authentication mechanisms.

These contributions collectively strengthen the security of EC and IoT systems, fostering a more resilient and sustainable architectural framework.

6. Discussion and Future Research Directions

Although EC was developed to address issues such as bandwidth limitations, latency, and real-time data processing arising from the integration with CC, there are still some fundamental challenges and unresolved research areas in this field, despite significant progress. In particular, the rapid increase in IoT devices has made scalability between these devices a necessity. However, managing this scalability within EC remains a major challenge. Moreover, the limited storage capacity of edge devices further increases the dependency on cloud computing. Therefore, the development of new communication protocols and the optimization of existing ones are essential to ensure interoperability between edge and cloud computing systems.

The advantage of EC in reducing latency presents a significant potential for developing effective architectures in critical processes, such as real-time control and monitoring in power systems. However, these processes require a more detailed investigation to develop applicable and scalable solutions. One key research challenge is optimizing the performance of connectivity layers within the EC architecture. Achieving high-speed data transfer and low-latency communication between hierarchical layers is fundamental to ensuring system efficiency and reliability. To establish a well-defined taxonomy, the EC architecture can be categorized into three primary layers: the Device-Edge Layer, the Edge-Server Layer, and the Cloud Layer. The Device-Edge Layer consists of resource-constrained IoT devices, sensors, and smart meters deployed at the network periphery. These devices perform preliminary data processing, local analytics, and real-time decision-making to reduce communication overhead and minimize latency. The computational capacity in this layer is typically limited, necessitating efficient workload distribution and lightweight data processing techniques. The Edge-Server Layer acts as an intermediary between the device layer and the cloud, comprising edge servers, gateways, and micro data centers with enhanced computational and storage capabilities. These nodes aggregate, filter, and process data received from multiple IoT devices before transmitting the refined information to the cloud. Additionally, this layer supports low-latency services and real-time analytics, enabling localized intelligence while reducing dependency on centralized cloud infrastructures. The Cloud Layer represents the centralized computing infrastructure responsible for deep learning, large-scale data analytics, and long-term storage. It provides extensive computational resources for complex model training, historical data analysis, and large-scale optimization. While CC ensures high processing power, it introduces latency and bandwidth constraints, making efficient coordination with the edge layers essential. The effectiveness of EC depends on the seamless interaction between these layers. Therefore, selecting appropriate networking technologies, optimizing communication protocols, and defining standardized roles for each layer are critical research directions. Future work should focus on refining

these hierarchical structures to enhance the performance, scalability, and interoperability of EC-enabled power systems.

With the shift in data processing to the network's edge, data privacy and security have become even more critical. In this context, developing and testing effective mechanisms to protect sensitive data is vital for EC to be adopted as a sustainable and reliable technology. In the future, new cryptographic approaches, security mechanisms, and hybrid architectures aimed at enhancing data privacy could significantly increase the effectiveness of EC. Additionally, network optimization, energy efficiency, and system resilience will remain significant research topics in the expanding application areas of EC.

Prediction systems based on AI techniques, including ML algorithms, hold significant potential to enhance the efficient utilization of EC in power systems and energy management processes.Furthermore, the integration of distributed security solutions, such as blockchain, with EC could play a key role in enhancing data security and integrity. By developing EC-based solutions, the next generation of power distribution networks can be made smarter, more autonomous, and more secure. The use of EC in energy management and power systems will not only improve operational and analytical processes but also contribute to making the overall energy ecosystem more efficient and sustainable.

The development of quantum-resistant encryption techniques is becoming increasingly crucial to secure sensitive data processed at the edge of EC systems. As quantum computing evolves, traditional cryptographic methods may become vulnerable, making it essential to integrate quantum-resistant algorithms into EC architectures. This will enhance the resilience of EC systems and protect against emerging cybersecurity threats, particularly in the context of smart grids.

In addition, the rise in energy-harvesting edge devices presents a significant opportunity to improve the sustainability and autonomy of EC systems. By harnessing energy from environmental sources such as solar, wind, or vibration, these devices reduce dependence on traditional power supplies. In smart grids, energy-harvesting technologies could enable self-sustaining edge nodes, enhancing the energy efficiency and reliability of the network, especially in remote or off-grid areas. Future research should focus on optimizing energy harvesting mechanisms to improve the scalability and resilience of EC systems in diverse environments.

7. Conclusions

EC has emerged as a transformative solution for modern power systems, addressing critical challenges associated with latency, bandwidth limitations, and real-time data processing. This comprehensive review has systematically examined the state-of-the-art EC applications in energy distribution systems, highlighting its role in optimizing grid operations and enhancing overall efficiency. The study explored various EC architectures, methodologies, and integration strategies with advanced technologies such as AI, machine learning, and IoT. Additionally, it examined key technological components, including containerization, software platforms, and hardware architectures, which play a crucial role in the deployment and efficiency of EC systems. Key applications of EC, including data transmission, monitoring and control, microgrid systems, smart metering, anomaly detection, energy management, and fault detection, have been analyzed in depth. The findings suggest that EC significantly improves system resilience, reliability, and operational flexibility by enabling local data processing and reducing dependency on centralized cloud infrastructures.

Despite these advantages, several challenges remain, such as security concerns, network optimization, and the need for scalable deployment strategies. Ensuring data privacy, developing robust security frameworks, and optimizing communication protocols are

crucial for the widespread adoption of EC in smart grids. Additionally, integrating EC with emerging technologies offers opportunities to enhance security, efficiency, and automation. Future research should focus on refining EC architectures, improving interoperability between EC and CC, and addressing real-time energy management constraints.

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Abbreviations

IoT	Internet of Things
LAN	Local Area Network
EC	Edge Computing
API	Application Programming Interface
GA	Genetic Algorithm
IED	Improvised Explosive Devices
CC	Cloud Computing
CI	Computational Intelligence
DMLS	Distributed Multi-level Storage
AI	Artificial Intelligence
MEC	Multi-access Edge Computing
WAN	Wide Area Network
FCN	Fog Computing Network
ILP	Integer Linear Programming
ML	Machine Learning
QoS	Quality of Service
MCC	Mobile Cloud Computing
QoE	Quality of Experience
AR	Augmented Reality
P2P	Peer-to-peer
NFV	Network Function Virtualisation
CVT	Container-based Virtualisation Technology
VM	Virtual Machine
CLI	Command Line Interface
JTSC	Joint Task Scheduling and Containerisation
BLE	Bluetooth Low Energy
CoAP	Constrained Application Protocol
MQTT	Minimum Message Queuing Telemetry Transport
SQL	Structured Query Language
TLS	Transport Layer Security
AMQP	Advanced Message Queuing Protocol
XMPP	Extensible Messaging and Presence Protocol
TCP	Transmission Control Protocol
AioT	AI of Things
TATSR	Time Average Transmission Success Rate

SG	Smart Grid
6LoWPAN	Low Power Personal Area Network
DER	Distributed Energy Resources
DG	Distributed Generators
SPVI	Smart PV-based Inverters
FDI	False Data Injection
SE	State Estimation
ANN	Artificial Neural Network
VVC	Voltage/VAR Control
IoS	Internet of Services
MADRL	Multi-agent Deep Reinforcement Learning
AMI	Advanced Metering Infrastructure
NILM	Non-intrusive Load Monitoring
MDMS	Meter Data Management System
HAN	Home Area Network
SM	Smart Metering
DNN	Deep Neural Network
SVR	Support Vector Regression
KNN	K-Nearest Neighbor
AMI	Advanced Metering Infrastructure
LV	Low Voltage
DTU	Distribution Transformer Unit
PMU	Phasor Measurement Data
DSSE	Distribution System State Estimation
DMS	Distribution Management System
VAR	Vector Autoregression
FDIA	False Data Injection Attack
HEMS	Home Energy Management System
EMS	Energy Management System
ECCREM	Edge-cloud Collaboration-based Residential Energy Management
SDN	Software-Defined Network
TATSR	Time Average Transmission Success Ratio
PD-IoT	Power Distribution Internet of Things
MDRL	Model-based Deep Reinforcement Learning
LoRa	Long Range
SCADA	Supervisory Control and Data Acquisition
VVC	Volt/var Control
PD-IoT	Power Distribution Internet of Things
FL	Federated Learning
DRL	Deep Reinforcement Learning
HDTG	Hierarchical Decision-making Strategy Based on Prediction Strategy and Task Grading
LoRa	Long Range
DT	Decision Tree
TWAM	Traveling Wave Acquisition Module
IIoT	Industrial Internet of Things
DT	Digital Twins
SLG	Single-Phase Ground
DRL	Deep Reinforcement Learning
CPS	Cyber-Physical Systems
RBF	Radial Basis Function

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